

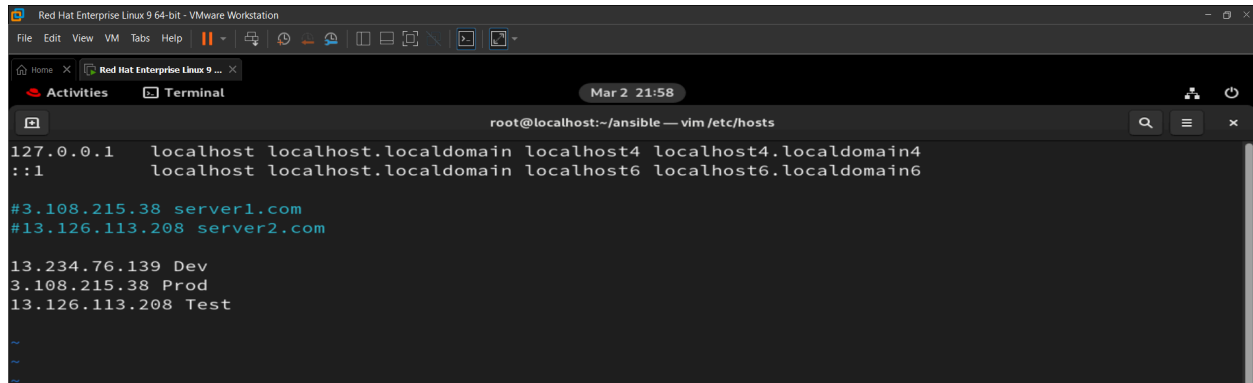
Name : Kartik Jain

TR:2

Ansible Assignment : 1

Q1. Configure Ansible and Managed Hosts:

- Set up Ansible on the control node.
- Configure three managed host machines: Dev, Prod, and Test.

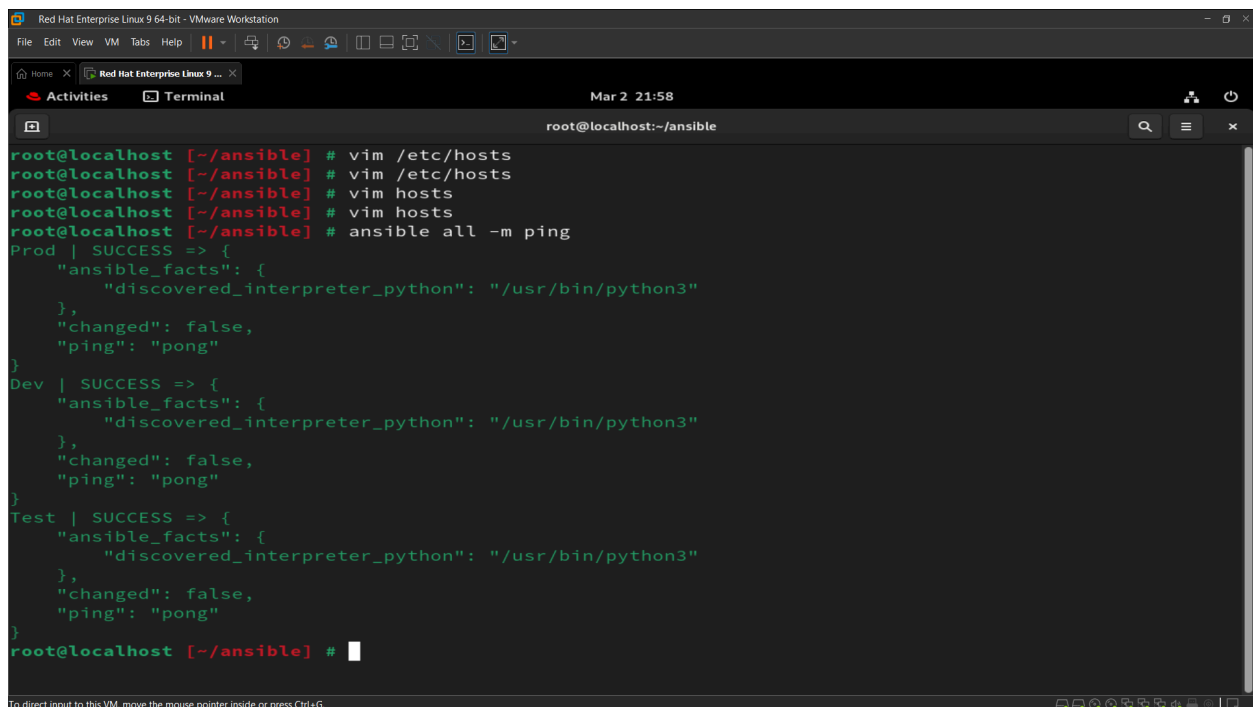


```
root@localhost:~/ansible — vim /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6

#3.108.215.38 server1.com
#13.126.113.208 server2.com

13.234.76.139 Dev
3.108.215.38 Prod
13.126.113.208 Test
```

- Verify connectivity all managed host machines: Dev, Prod, and Test.

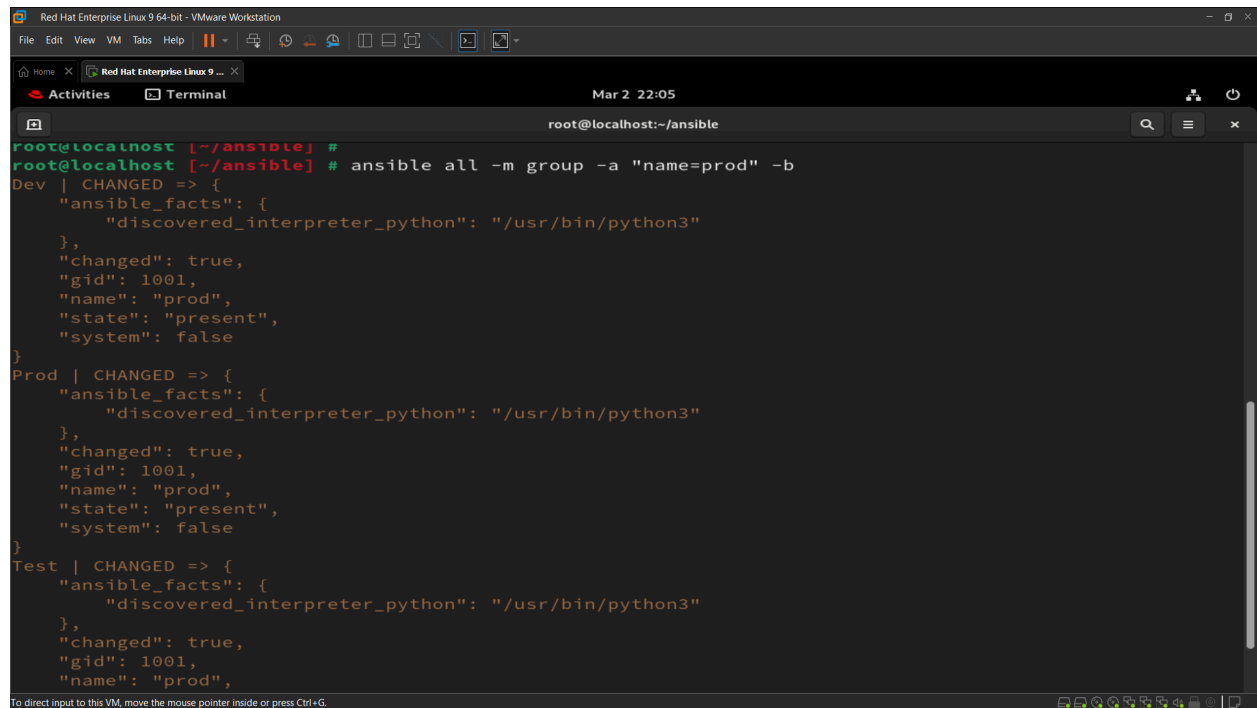


```
root@localhost [~/ansible] # vim /etc/hosts
root@localhost [~/ansible] # vim /etc/hosts
root@localhost [~/ansible] # vim hosts
root@localhost [~/ansible] # vim hosts
root@localhost [~/ansible] # ansible all -m ping
Prod | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
Dev | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
Test | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
root@localhost [~/ansible] #
```

Q2. User and Group Creation:

→ Create three users: Ramesh, Priya, and Suresh, each with their respective primary groups:

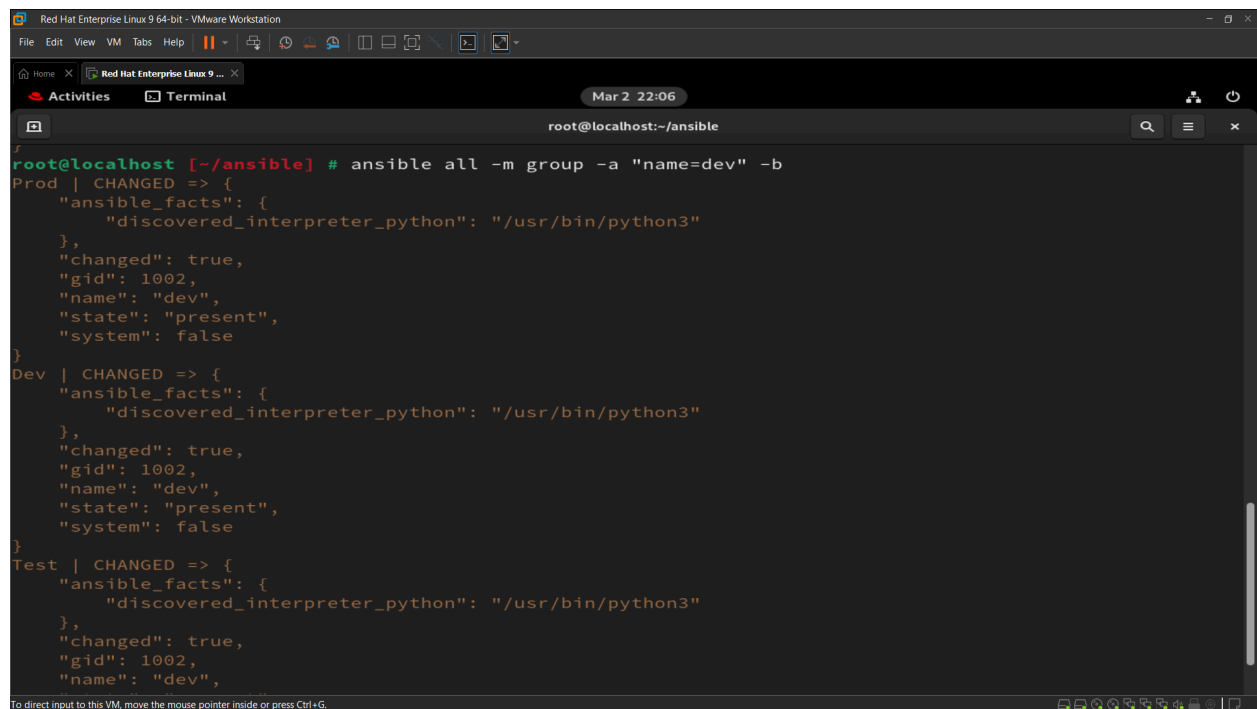
Group : prod



A terminal window titled "Red Hat Enterprise Linux 9 64-bit - VMware Workstation" showing an Ansible command and its output. The command is `ansible all -m group -a "name=prod" -b`. The output shows the configuration for the 'prod' group, including its name, gid (1001), state (present), and system (false). The terminal also shows the configuration for the 'dev' and 'test' groups, which are identical to 'prod'.

```
root@localhost [~/ansible] #  
root@localhost [~/ansible] # ansible all -m group -a "name=prod" -b  
Dev | CHANGED => {  
  "ansible_facts": {  
    "discovered_interpreter_python": "/usr/bin/python3"  
  },  
  "changed": true,  
  "gid": 1001,  
  "name": "prod",  
  "state": "present",  
  "system": false  
}  
Prod | CHANGED => {  
  "ansible_facts": {  
    "discovered_interpreter_python": "/usr/bin/python3"  
  },  
  "changed": true,  
  "gid": 1001,  
  "name": "prod",  
  "state": "present",  
  "system": false  
}  
Test | CHANGED => {  
  "ansible_facts": {  
    "discovered_interpreter_python": "/usr/bin/python3"  
  },  
  "changed": true,  
  "gid": 1001,  
  "name": "prod",  
  "state": "present",  
  "system": false  
}
```

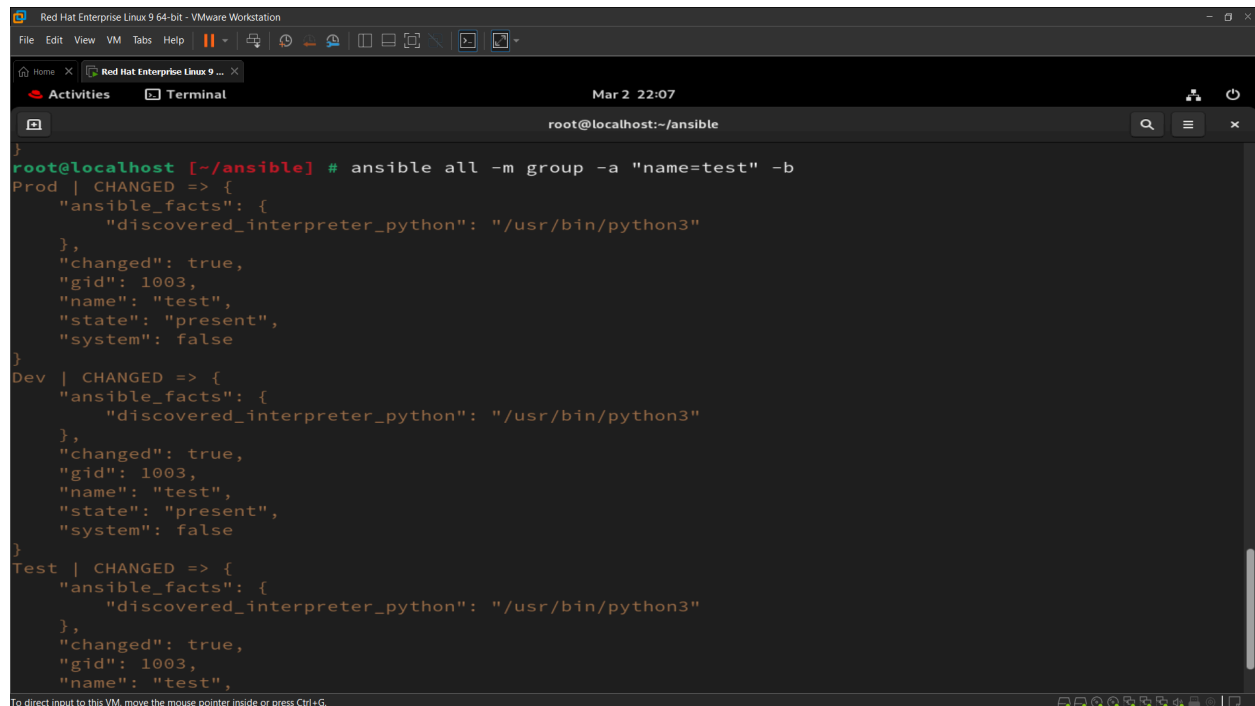
Group : dev



A terminal window titled "Red Hat Enterprise Linux 9 64-bit - VMware Workstation" showing an Ansible command and its output. The command is `ansible all -m group -a "name=dev" -b`. The output shows the configuration for the 'dev' group, including its name, gid (1002), state (present), and system (false). The terminal also shows the configuration for the 'prod' and 'test' groups, which are identical to 'dev'.

```
root@localhost [~/ansible] # ansible all -m group -a "name=dev" -b  
Prod | CHANGED => {  
  "ansible_facts": {  
    "discovered_interpreter_python": "/usr/bin/python3"  
  },  
  "changed": true,  
  "gid": 1002,  
  "name": "dev",  
  "state": "present",  
  "system": false  
}  
Dev | CHANGED => {  
  "ansible_facts": {  
    "discovered_interpreter_python": "/usr/bin/python3"  
  },  
  "changed": true,  
  "gid": 1002,  
  "name": "dev",  
  "state": "present",  
  "system": false  
}  
Test | CHANGED => {  
  "ansible_facts": {  
    "discovered_interpreter_python": "/usr/bin/python3"  
  },  
  "changed": true,  
  "gid": 1002,  
  "name": "dev",  
  "state": "present",  
  "system": false  
}
```

Group : test

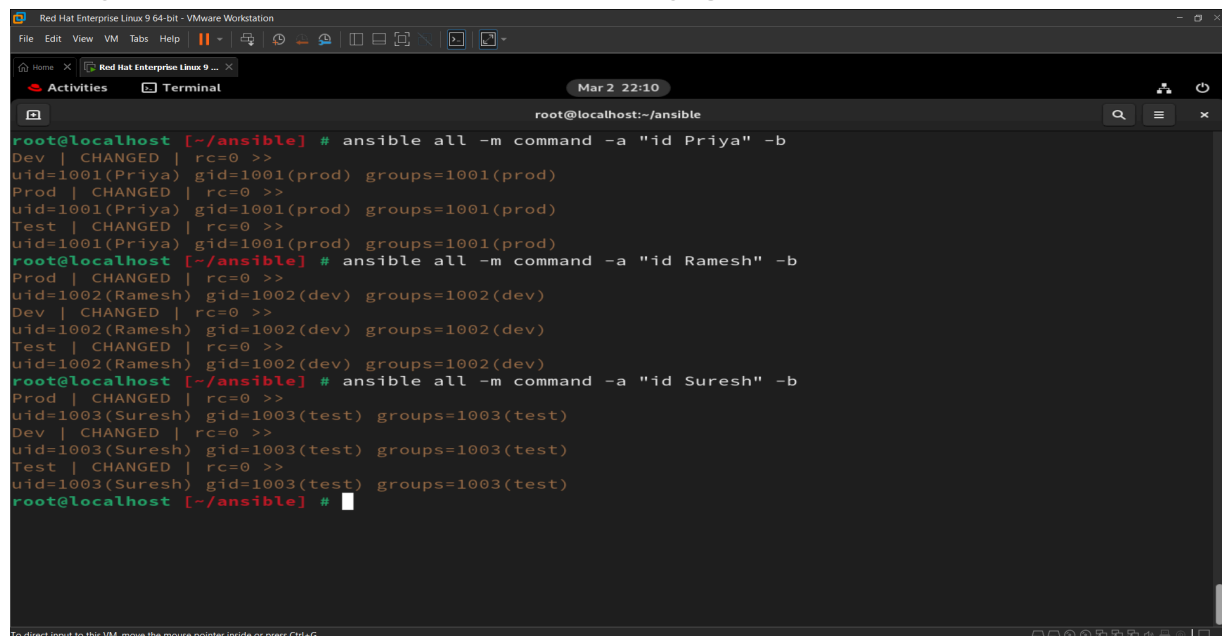


A terminal window titled "Red Hat Enterprise Linux 9 64-bit - VMware Workstation" showing an Ansible command and its output. The command is `ansible all -m group -a "name=test" -b`. The output shows the configuration for the 'test' group across three hosts: Prod, Dev, and Test. Each host has the same configuration: `ansible_facts: {discovered_interpreter_python: "/usr/bin/python3"}`, `changed: true`, `gid: 1003`, `name: "test"`, `state: "present"`, and `system: false`.

```
root@localhost [~/ansible] # ansible all -m group -a "name=test" -b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 1003,
  "name": "test",
  "state": "present",
  "system": false
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 1003,
  "name": "test",
  "state": "present",
  "system": false
}
Test | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 1003,
  "name": "test",
  "state": "present",
  "system": false
}
```

1. Priya: Primary group prod
2. Ramesh: Primary group dev
3. Suresh: Primary group test

• Verify all user with respective primary group prod,dev,test



A terminal window titled "Red Hat Enterprise Linux 9 64-bit - VMware Workstation" showing three Ansible commands to verify user group assignments. The first command is `ansible all -m command -a "id Priya" -b`, the second is `ansible all -m command -a "id Ramesh" -b`, and the third is `ansible all -m command -a "id Suresh" -b`. The output shows the user details for each user across the three hosts.

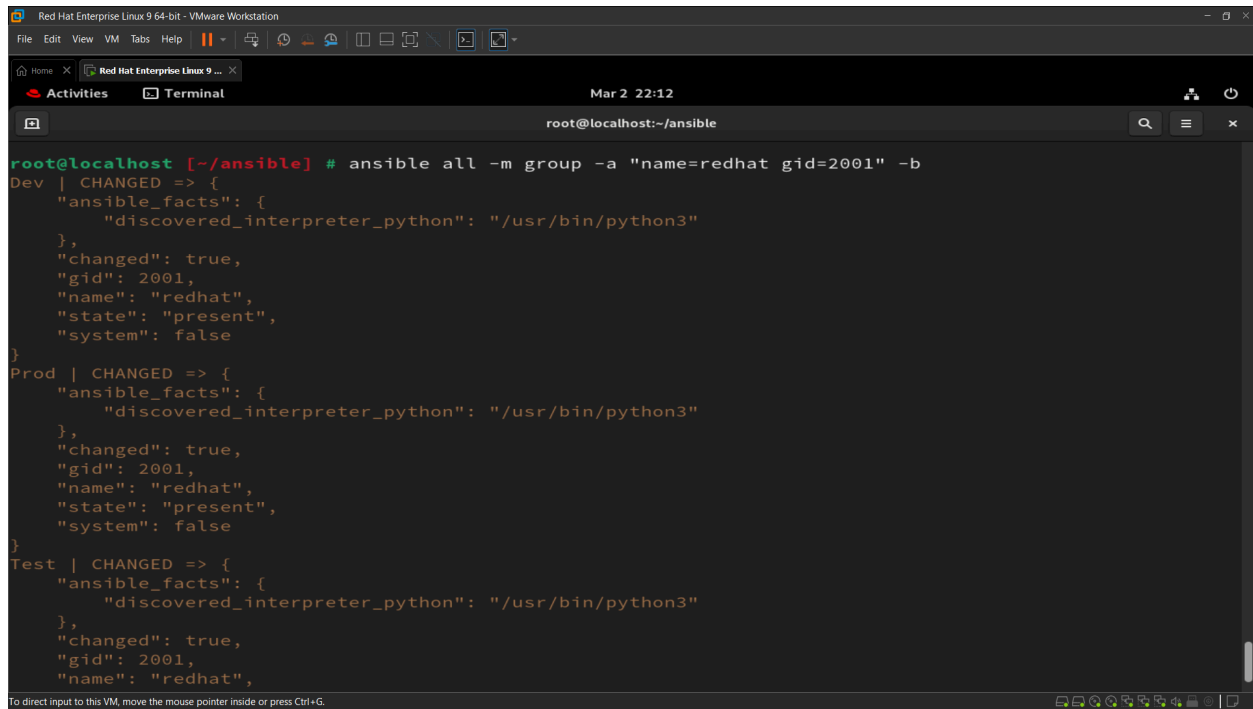
```
root@localhost [~/ansible] # ansible all -m command -a "id Priya" -b
Dev | CHANGED | rc=0 >>
uid=1001(Priya) gid=1001(prod) groups=1001(prod)
Prod | CHANGED | rc=0 >>
uid=1001(Priya) gid=1001(prod) groups=1001(prod)
Test | CHANGED | rc=0 >>
uid=1001(Priya) gid=1001(prod) groups=1001(prod)
root@localhost [~/ansible] # ansible all -m command -a "id Ramesh" -b
Prod | CHANGED | rc=0 >>
uid=1002(Ramesh) gid=1002(dev) groups=1002(dev)
Dev | CHANGED | rc=0 >>
uid=1002(Ramesh) gid=1002(dev) groups=1002(dev)
Test | CHANGED | rc=0 >>
uid=1002(Ramesh) gid=1002(dev) groups=1002(dev)
root@localhost [~/ansible] # ansible all -m command -a "id Suresh" -b
Prod | CHANGED | rc=0 >>
uid=1003(Suresh) gid=1003(test) groups=1003(test)
Dev | CHANGED | rc=0 >>
uid=1003(Suresh) gid=1003(test) groups=1003(test)
Test | CHANGED | rc=0 >>
uid=1003(Suresh) gid=1003(test) groups=1003(test)
root@localhost [~/ansible] #
```

- Create two secondary groups: redhat and ubuntu with the following group IDs:

- redhat (Group ID: 2001)

- ubuntu (Group ID: 2002)

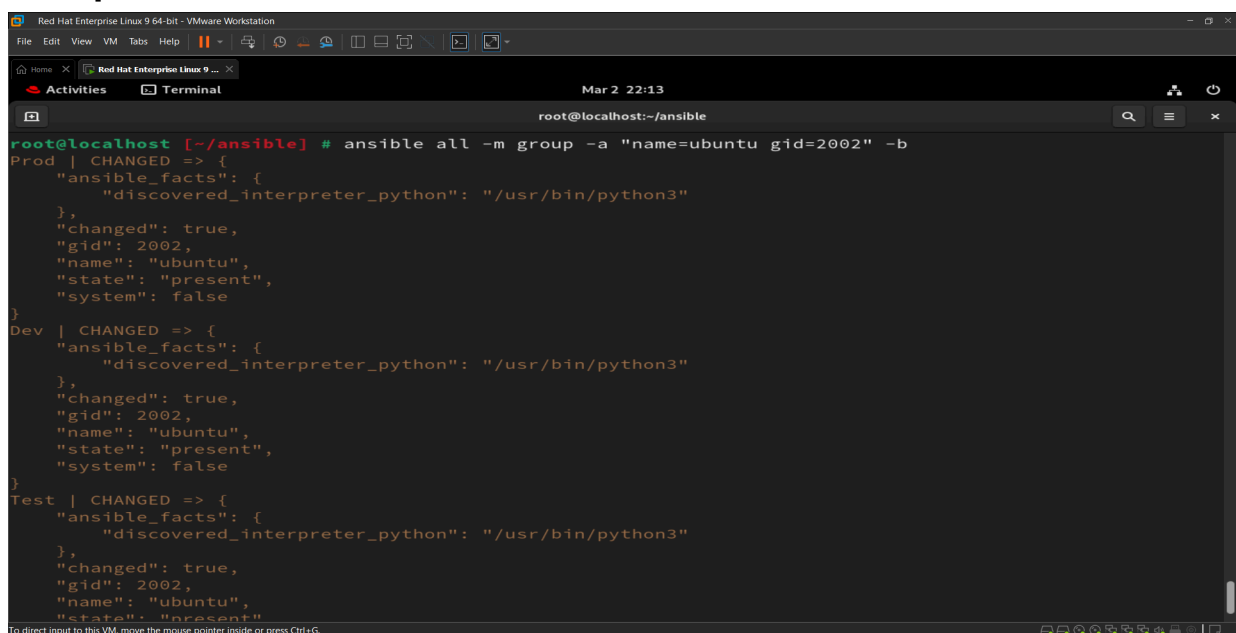
Group : redhat



A terminal window titled "Red Hat Enterprise Linux 9 64-bit - VMware Workstation" showing an Ansible command and its output. The command is `ansible all -m group -a "name=redhat gid=2001" -b`. The output shows the command being executed on three hosts: Dev, Prod, and Test. Each host shows a "CHANGED" status and a JSON output indicating the group was created successfully with the specified name and ID.

```
root@localhost:~/ansible # ansible all -m group -a "name=redhat gid=2001" -b
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 2001,
  "name": "redhat",
  "state": "present",
  "system": false
}
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 2001,
  "name": "redhat",
  "state": "present",
  "system": false
}
Test | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 2001,
  "name": "redhat",
  "state": "present",
  "system": false
}
```

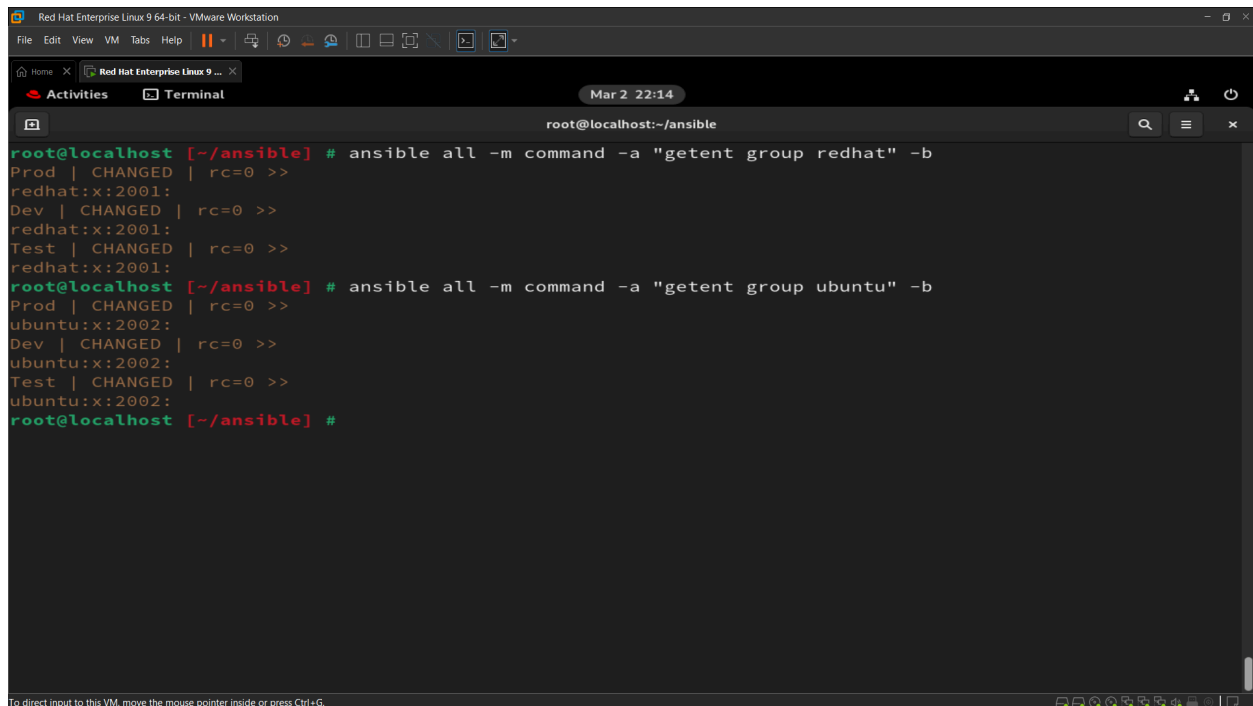
Group: ubuntu



A terminal window titled "Red Hat Enterprise Linux 9 64-bit - VMware Workstation" showing an Ansible command and its output. The command is `ansible all -m group -a "name=ubuntu gid=2002" -b`. The output shows the command being executed on three hosts: Prod, Dev, and Test. Each host shows a "CHANGED" status and a JSON output indicating the group was created successfully with the specified name and ID.

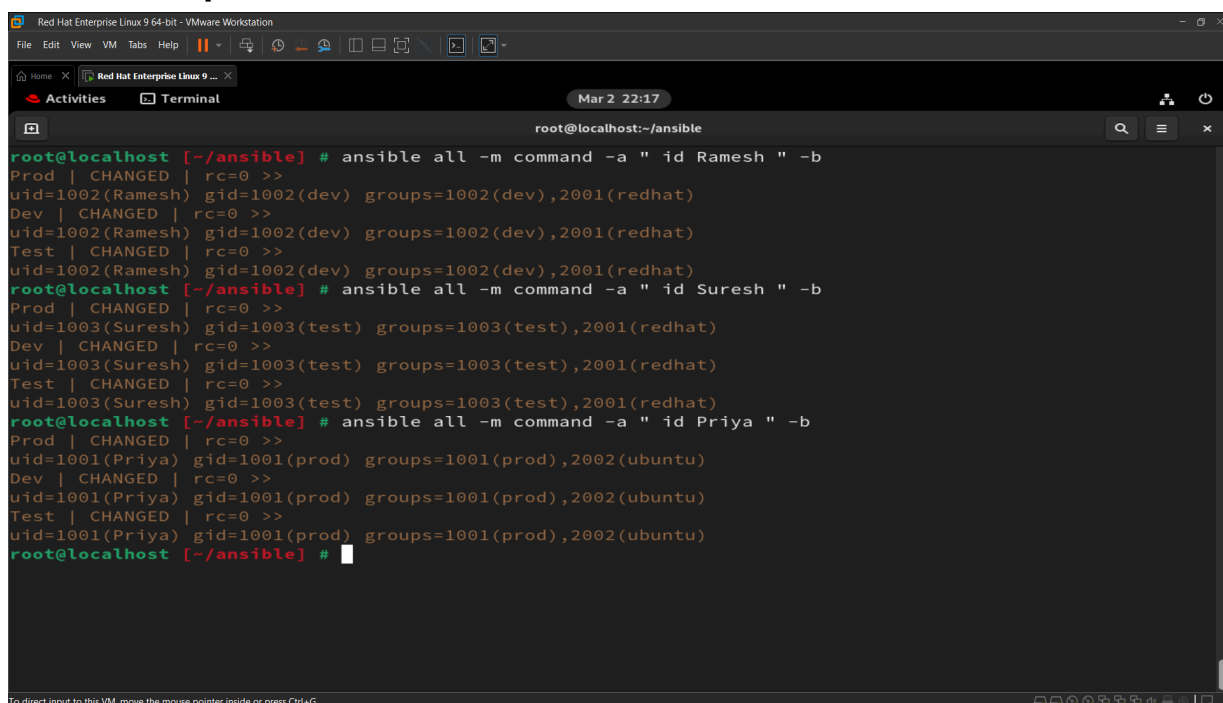
```
root@localhost:~/ansible # ansible all -m group -a "name=ubuntu gid=2002" -b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 2002,
  "name": "ubuntu",
  "state": "present",
  "system": false
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 2002,
  "name": "ubuntu",
  "state": "present",
  "system": false
}
Test | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 2002,
  "name": "ubuntu",
  "state": "present",
  "system": false
}
```

- **Verify group with gid :**



```
root@localhost: ~/ansible
root@localhost [~/ansible] # ansible all -m command -a "getent group redhat" -b
Prod | CHANGED | rc=0 >>
redhat:x:2001:
Dev | CHANGED | rc=0 >>
redhat:x:2001:
Test | CHANGED | rc=0 >>
redhat:x:2001:
root@localhost [~/ansible] # ansible all -m command -a "getent group ubuntu" -b
Prod | CHANGED | rc=0 >>
ubuntu:x:2002:
Dev | CHANGED | rc=0 >>
ubuntu:x:2002:
Test | CHANGED | rc=0 >>
ubuntu:x:2002:
root@localhost [~/ansible] #
```

- **Assign secondary group memberships:**
- **Ramesh and Suresh → Secondary group: redhat**
- **Priya → Secondary group: ubuntu**
- **Verify secondary group memberships.**

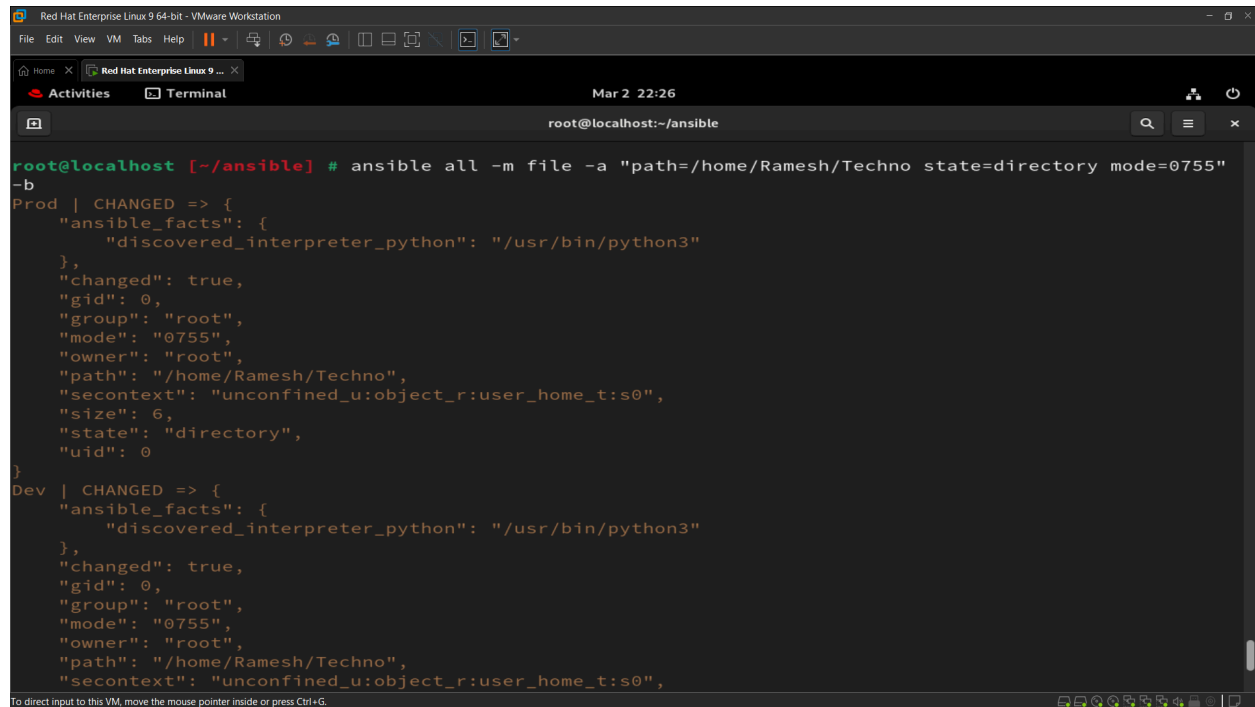


```
root@localhost: ~/ansible
root@localhost [~/ansible] # ansible all -m command -a " id Ramesh " -b
Prod | CHANGED | rc=0 >>
uid=1002(Ramesh) gid=1002(dev) groups=1002(dev),2001(redhat)
Dev | CHANGED | rc=0 >>
uid=1002(Ramesh) gid=1002(dev) groups=1002(dev),2001(redhat)
Test | CHANGED | rc=0 >>
uid=1002(Ramesh) gid=1002(dev) groups=1002(dev),2001(redhat)
root@localhost [~/ansible] # ansible all -m command -a " id Suresh " -b
Prod | CHANGED | rc=0 >>
uid=1003(Suresh) gid=1003(test) groups=1003(test),2001(redhat)
Dev | CHANGED | rc=0 >>
uid=1003(Suresh) gid=1003(test) groups=1003(test),2001(redhat)
Test | CHANGED | rc=0 >>
uid=1003(Suresh) gid=1003(test) groups=1003(test),2001(redhat)
root@localhost [~/ansible] # ansible all -m command -a " id Priya " -b
Prod | CHANGED | rc=0 >>
uid=1001(Priya) gid=1001(prod) groups=1001(prod),2002(ubuntu)
Dev | CHANGED | rc=0 >>
uid=1001(Priya) gid=1001(prod) groups=1001(prod),2002(ubuntu)
Test | CHANGED | rc=0 >>
uid=1001(Priya) gid=1001(prod) groups=1001(prod),2002(ubuntu)
root@localhost [~/ansible] #
```

Q3: Directory and File Tasks:

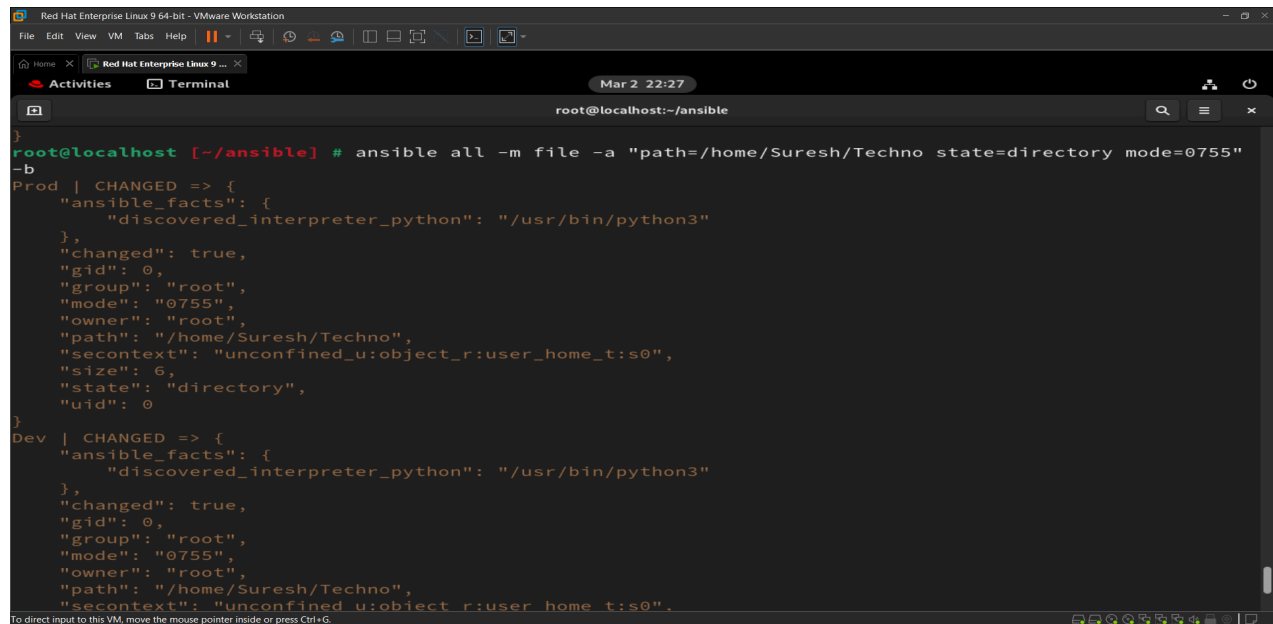
- Create a directory named Techno in the home directories of all three users.

Ramesh :



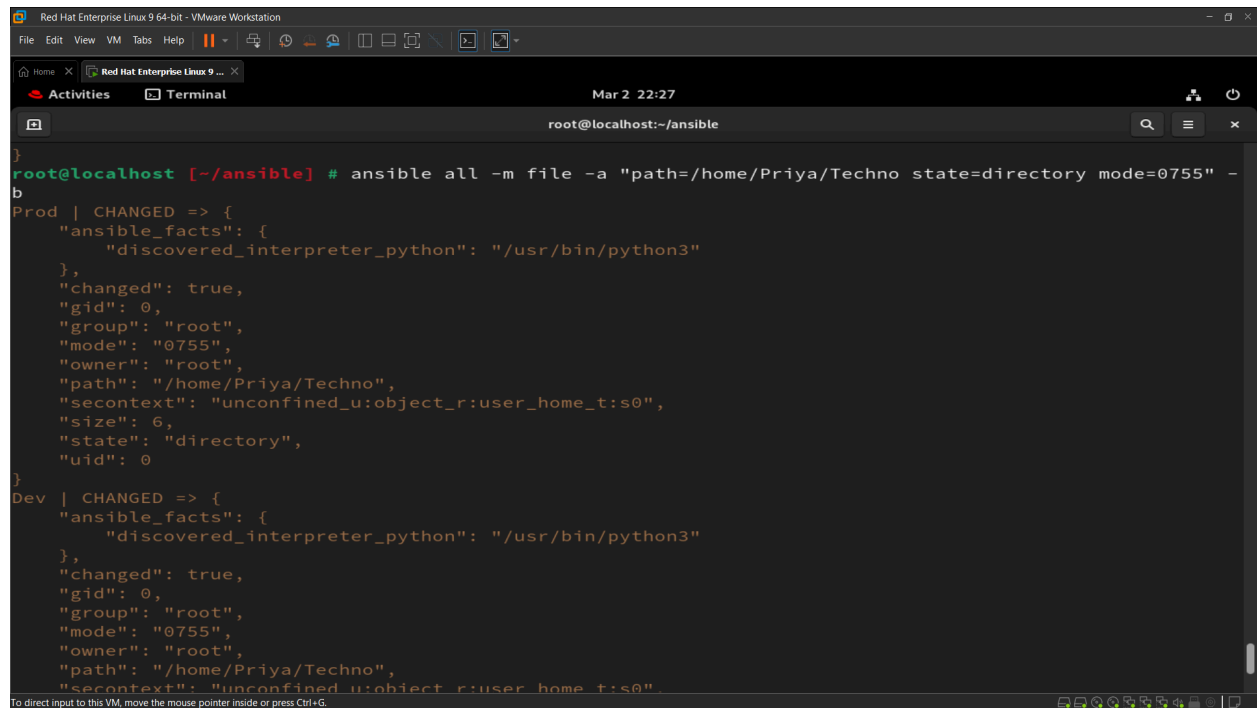
```
root@localhost [~/ansible] # ansible all -m file -a "path=/home/Ramesh/Techno state=directory mode=0755"
-b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 0,
  "group": "root",
  "mode": "0755",
  "owner": "root",
  "path": "/home/Ramesh/Techno",
  "secontext": "unconfined_u:object_r:user_home_t:s0",
  "size": 6,
  "state": "directory",
  "uid": 0
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 0,
  "group": "root",
  "mode": "0755",
  "owner": "root",
  "path": "/home/Ramesh/Techno",
  "secontext": "unconfined_u:object_r:user_home_t:s0",
  "size": 6,
  "state": "directory",
  "uid": 0
}
```

Suresh :



```
root@localhost [~/ansible] # ansible all -m file -a "path=/home/Suresh/Techno state=directory mode=0755"
-b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 0,
  "group": "root",
  "mode": "0755",
  "owner": "root",
  "path": "/home/Suresh/Techno",
  "secontext": "unconfined_u:object_r:user_home_t:s0",
  "size": 6,
  "state": "directory",
  "uid": 0
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 0,
  "group": "root",
  "mode": "0755",
  "owner": "root",
  "path": "/home/Suresh/Techno",
  "secontext": "unconfined_u:object_r:user_home_t:s0",
  "size": 6,
  "state": "directory",
  "uid": 0
}
```

Priya:

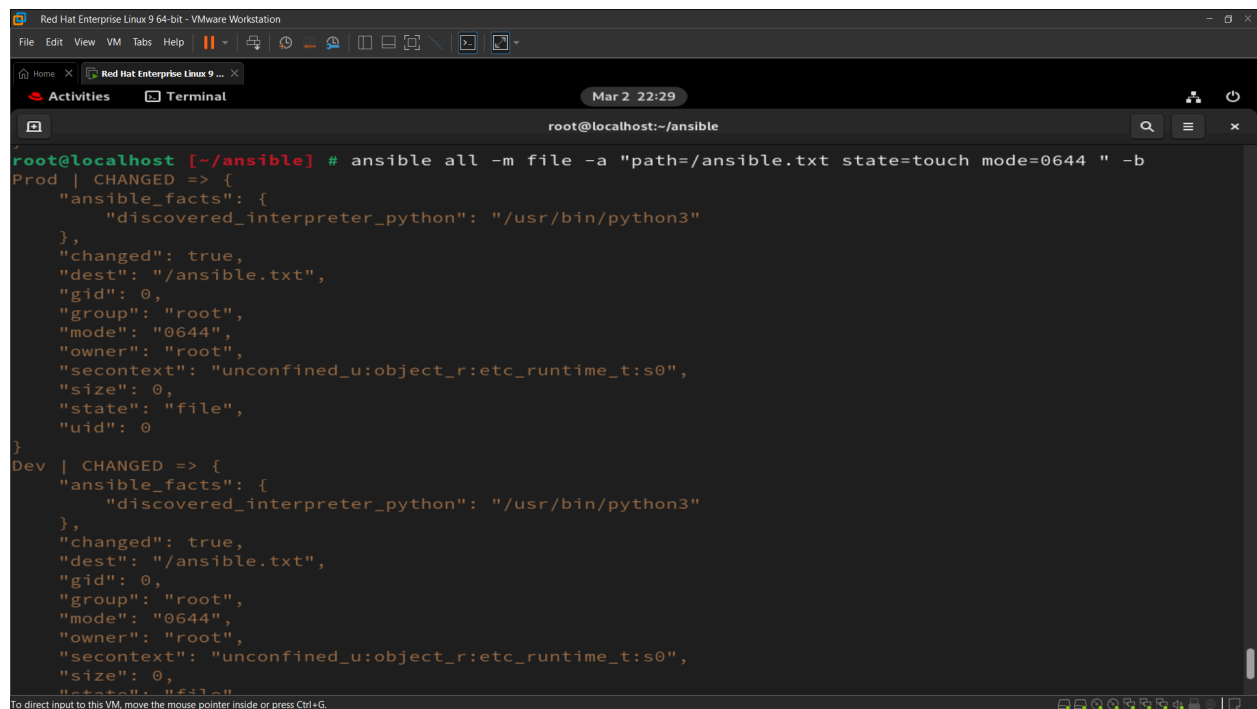


A terminal window titled "Red Hat Enterprise Linux 9 64-bit - VMware Workstation" showing an Ansible command and its output. The command is `ansible all -m file -a "path=/home/Priya/Techno state=directory mode=0755" -b`. The output shows the command was successful for both 'Prod' and 'Dev' hosts, creating the directory `/home/Priya/Techno` with mode `0755`. The output for 'Prod' is as follows:

```
}
root@localhost [~/ansible] # ansible all -m file -a "path=/home/Priya/Techno state=directory mode=0755" -b
b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 0,
  "group": "root",
  "mode": "0755",
  "owner": "root",
  "path": "/home/Priya/Techno",
  "secontext": "unconfined_u:object_r:user_home_t:s0",
  "size": 6,
  "state": "directory",
  "uid": 0
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "gid": 0,
  "group": "root",
  "mode": "0755",
  "owner": "root",
  "path": "/home/Priya/Techno",
  "secontext": "unconfined_u:object_r:user_home_t:s0"
}
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

- Create a file named `ansible.txt` in the root directory (/).

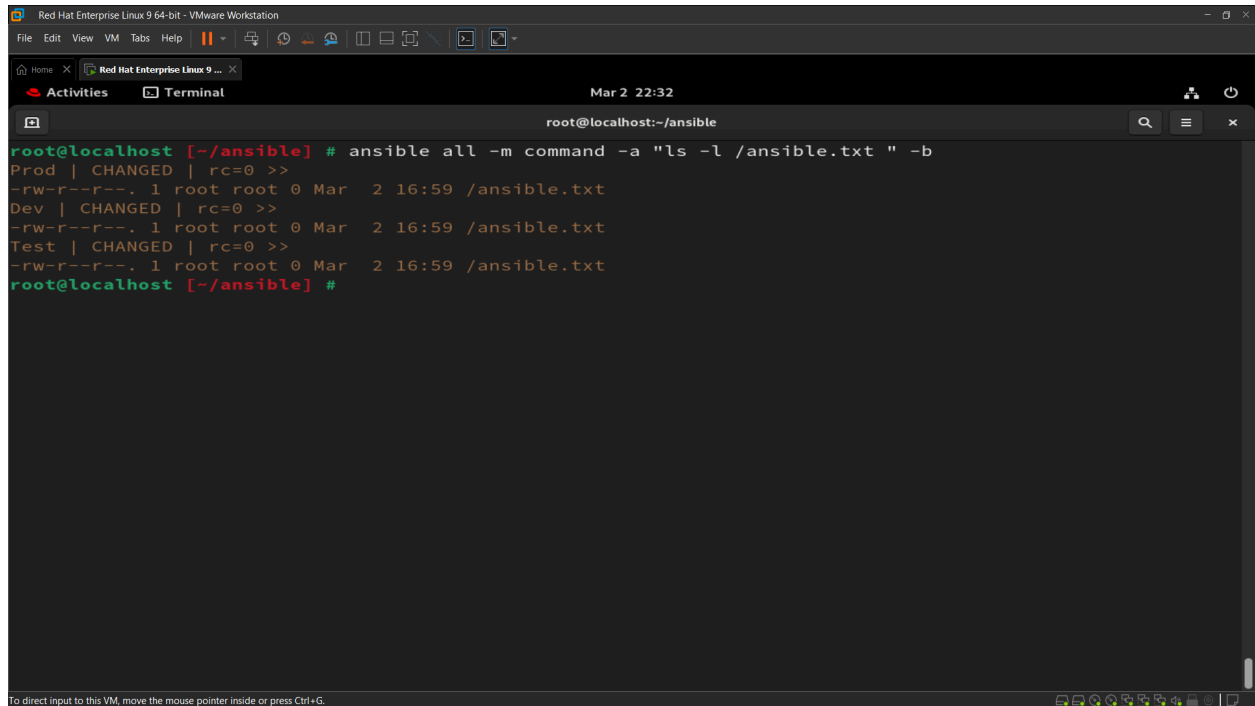


A terminal window titled "Red Hat Enterprise Linux 9 64-bit - VMware Workstation" showing an Ansible command and its output. The command is `ansible all -m file -a "path=/ansible.txt state=touch mode=0644" -b`. The output shows the command was successful for both 'Prod' and 'Dev' hosts, creating the file `/ansible.txt` with mode `0644`. The output for 'Prod' is as follows:

```
root@localhost [~/ansible] # ansible all -m file -a "path=/ansible.txt state=touch mode=0644" -b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "dest": "/ansible.txt",
  "gid": 0,
  "group": "root",
  "mode": "0644",
  "owner": "root",
  "secontext": "unconfined_u:object_r:etc_runtime_t:s0",
  "size": 0,
  "state": "file",
  "uid": 0
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "dest": "/ansible.txt",
  "gid": 0,
  "group": "root",
  "mode": "0644",
  "owner": "root",
  "secontext": "unconfined_u:object_r:etc_runtime_t:s0",
  "size": 0,
  "state": "file",
  "uid": 0
}
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

- Verify a file named ansible.txt in the root directory (/).

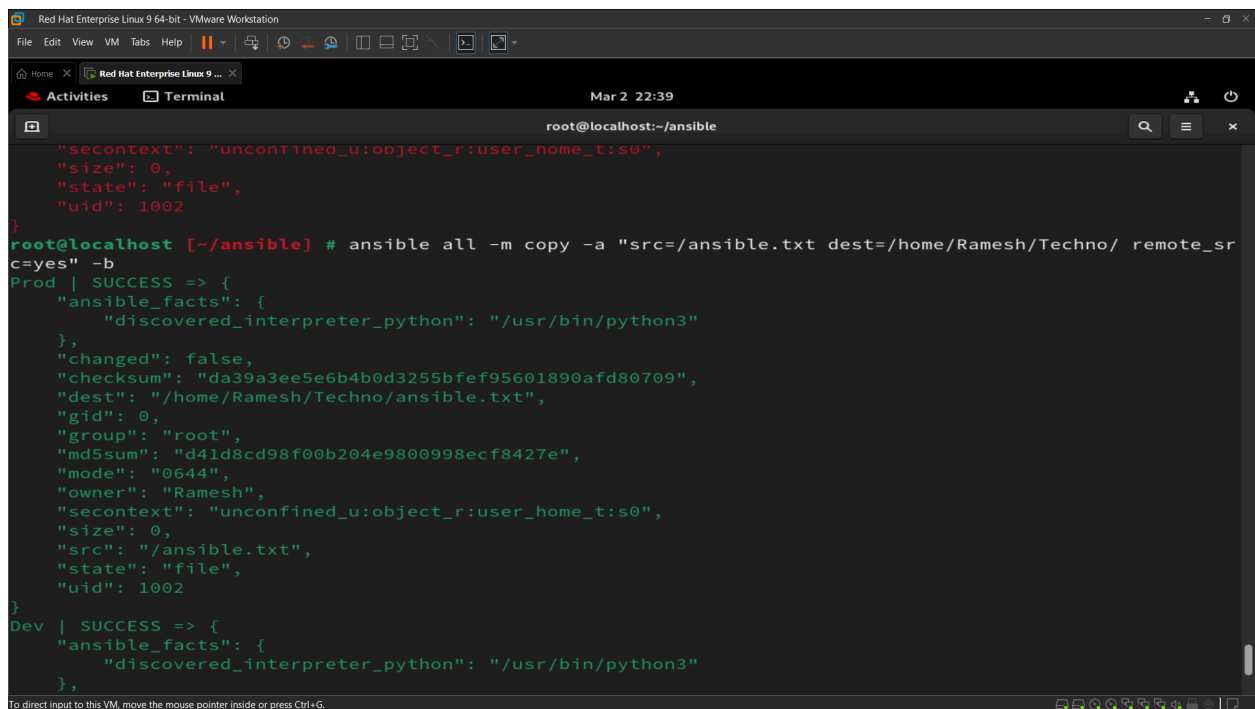


```
Red Hat Enterprise Linux 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
Activities Terminal Mar 2 22:32
root@localhost:~/ansible

root@localhost [~/ansible] # ansible all -m command -a "ls -l /ansible.txt" -b
Prod | CHANGED | rc=0 >>
-rw-r--r--. 1 root root 0 Mar  2 16:59 /ansible.txt
Dev | CHANGED | rc=0 >>
-rw-r--r--. 1 root root 0 Mar  2 16:59 /ansible.txt
Test | CHANGED | rc=0 >>
-rw-r--r--. 1 root root 0 Mar  2 16:59 /ansible.txt
root@localhost [~/ansible] #
```

- Copy the ansible.txt file into the /home/username/Techno directory of each user.

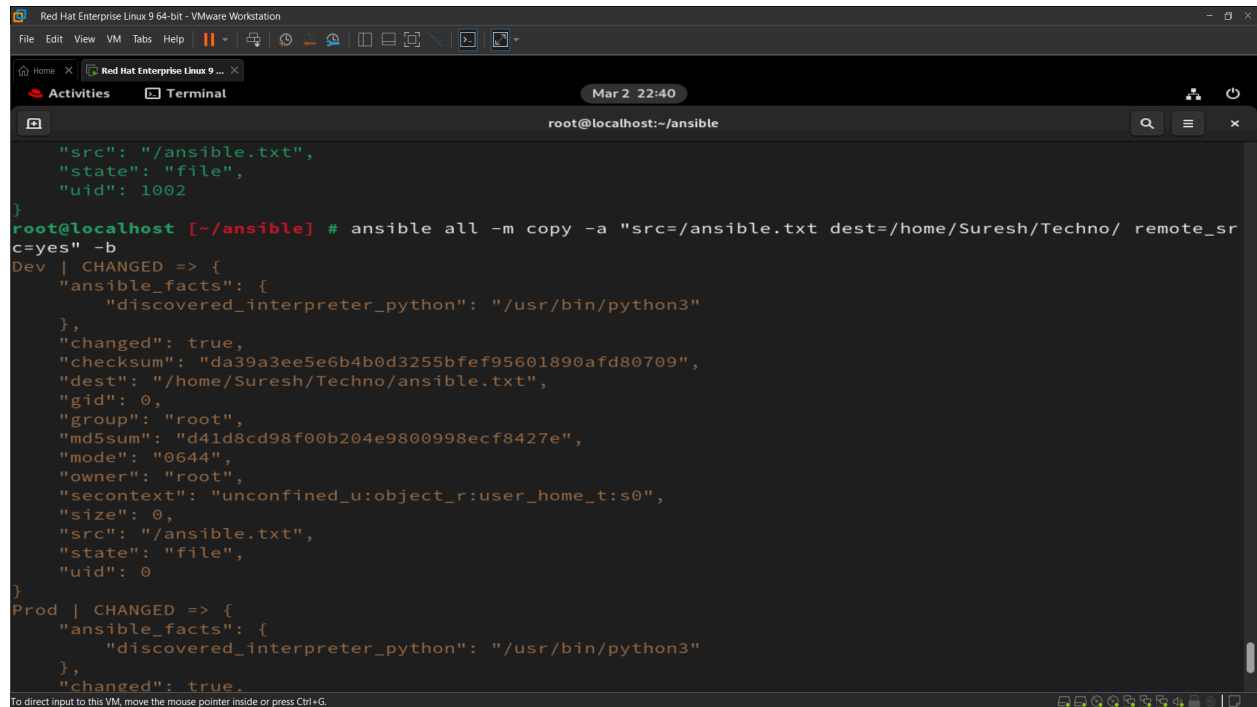
Ramesh:



```
Red Hat Enterprise Linux 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
Activities Terminal Mar 2 22:39
root@localhost:~/ansible

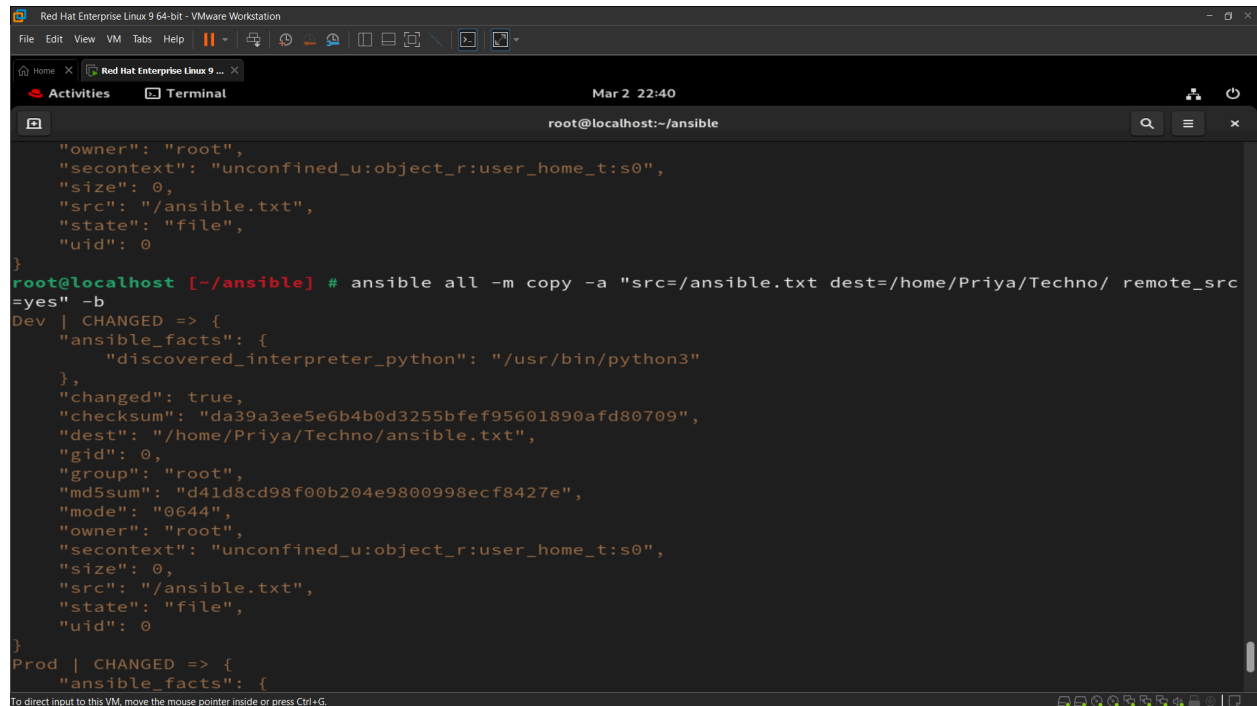
"secontext": "unconfined_u:object_r:user_home_t:s0",
"size": 0,
"state": "file",
"uid": 1002
}
root@localhost [~/ansible] # ansible all -m copy -a "src=/ansible.txt dest=/home/Ramesh/Techno/ remote_sr
c=yes" -b
Prod | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "checksum": "da39a3ee5e6b4b0d3255bfef95601890afd80709",
  "dest": "/home/Ramesh/Techno/ansible.txt",
  "gid": 0,
  "group": "root",
  "md5sum": "d41d8cd98f00b204e9800998ecf8427e",
  "mode": "0644",
  "owner": "Ramesh",
  "secontext": "unconfined_u:object_r:user_home_t:s0",
  "size": 0,
  "src": "/ansible.txt",
  "state": "file",
  "uid": 1002
}
Dev | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "checksum": "da39a3ee5e6b4b0d3255bfef95601890afd80709",
  "dest": "/home/Ramesh/Techno/ansible.txt",
  "gid": 0,
  "group": "root",
  "md5sum": "d41d8cd98f00b204e9800998ecf8427e",
  "mode": "0644",
  "owner": "Ramesh",
  "secontext": "unconfined_u:object_r:user_home_t:s0",
  "size": 0,
  "src": "/ansible.txt",
  "state": "file",
  "uid": 1002
}
root@localhost [~/ansible] #
```

Suresh :



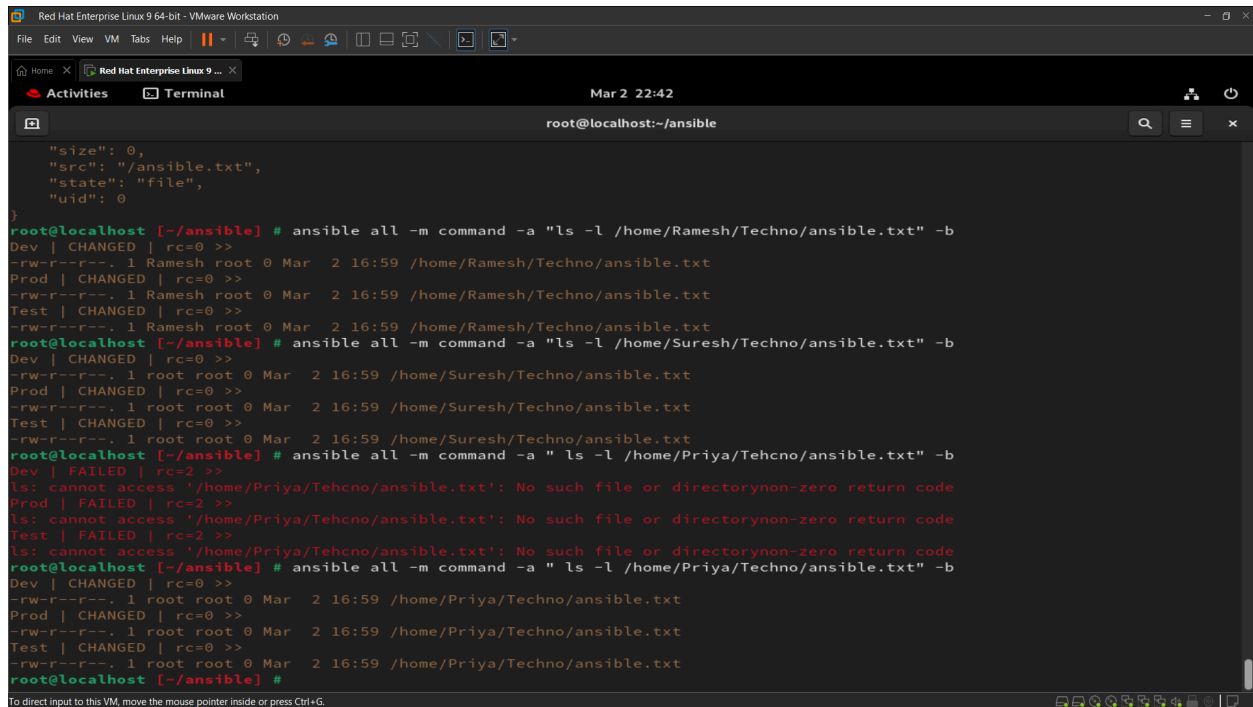
```
root@localhost:~/ansible
"src": "/ansible.txt",
"state": "file",
"uid": 1002
}
root@localhost [~/ansible] # ansible all -m copy -a "src=/ansible.txt dest=/home/Suresh/Techno/ remote_src=yes" -b
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "checksum": "da39a3ee5e6b4b0d3255bfef95601890afd80709",
  "dest": "/home/Suresh/Techno/ansible.txt",
  "gid": 0,
  "group": "root",
  "md5sum": "d41d8cd98f00b204e9800998ecf8427e",
  "mode": "0644",
  "owner": "root",
  "secontext": "unconfined_u:object_r:user_home_t:s0",
  "size": 0,
  "src": "/ansible.txt",
  "state": "file",
  "uid": 0
}
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
```

Priya :



```
root@localhost:~/ansible
"owner": "root",
"secontext": "unconfined_u:object_r:user_home_t:s0",
"size": 0,
"src": "/ansible.txt",
"state": "file",
"uid": 0
}
root@localhost [~/ansible] # ansible all -m copy -a "src=/ansible.txt dest=/home/Priya/Techno/ remote_src=yes" -b
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "checksum": "da39a3ee5e6b4b0d3255bfef95601890afd80709",
  "dest": "/home/Priya/Techno/ansible.txt",
  "gid": 0,
  "group": "root",
  "md5sum": "d41d8cd98f00b204e9800998ecf8427e",
  "mode": "0644",
  "owner": "root",
  "secontext": "unconfined_u:object_r:user_home_t:s0",
  "size": 0,
  "src": "/ansible.txt",
  "state": "file",
  "uid": 0
}
Prod | CHANGED => {
  "ansible_facts": {
```

- Verify the ansible.txt file into the /home/username/Techno directory of each user.

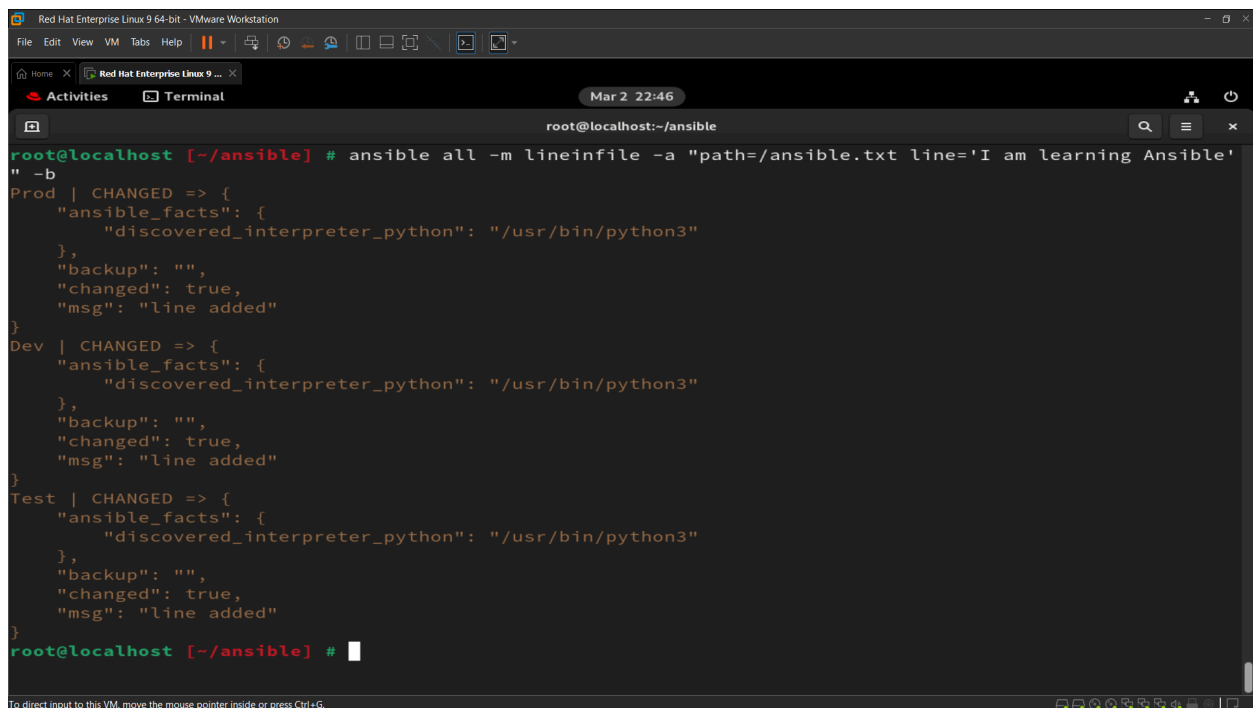


```

"size": 0,
"src": "/ansible.txt",
"state": "file",
"uid": 0
}
root@localhost [~/ansible] # ansible all -m command -a "ls -l /home/Ramesh/Techno/ansible.txt" -b
Dev | CHANGED | rc=0 >>
-rw-r--r--. 1 Ramesh root 0 Mar  2 16:59 /home/Ramesh/Techno/ansible.txt
Prod | CHANGED | rc=0 >>
-rw-r--r--. 1 Ramesh root 0 Mar  2 16:59 /home/Ramesh/Techno/ansible.txt
Test | CHANGED | rc=0 >>
-rw-r--r--. 1 Ramesh root 0 Mar  2 16:59 /home/Ramesh/Techno/ansible.txt
root@localhost [~/ansible] # ansible all -m command -a "ls -l /home/Suresh/Techno/ansible.txt" -b
Dev | CHANGED | rc=0 >>
-rw-r--r--. 1 root root 0 Mar  2 16:59 /home/Suresh/Techno/ansible.txt
Prod | CHANGED | rc=0 >>
-rw-r--r--. 1 root root 0 Mar  2 16:59 /home/Suresh/Techno/ansible.txt
Test | CHANGED | rc=0 >>
-rw-r--r--. 1 root root 0 Mar  2 16:59 /home/Suresh/Techno/ansible.txt
root@localhost [~/ansible] # ansible all -m command -a "ls -l /home/Priya/Techno/ansible.txt" -b
Dev | FAILED | rc=2 >>
[!s: cannot access '/home/Priya/Techno/ansible.txt': No such file or directorynon-zero return code
Prod | FAILED | rc=2 >>
[!s: cannot access '/home/Priya/Techno/ansible.txt': No such file or directorynon-zero return code
Test | FAILED | rc=2 >>
[!s: cannot access '/home/Priya/Techno/ansible.txt': No such file or directorynon-zero return code
root@localhost [~/ansible] # ansible all -m command -a "ls -l /home/Priya/Techno/ansible.txt" -b
Dev | CHANGED | rc=0 >>
-rw-r--r--. 1 root root 0 Mar  2 16:59 /home/Priya/Techno/ansible.txt
Prod | CHANGED | rc=0 >>
-rw-r--r--. 1 root root 0 Mar  2 16:59 /home/Priya/Techno/ansible.txt
Test | CHANGED | rc=0 >>
-rw-r--r--. 1 root root 0 Mar  2 16:59 /home/Priya/Techno/ansible.txt
root@localhost [~/ansible] #

```

- Add the line "I am learning Ansible" to the ansible.txt file using an Ansible ad-hoc command.

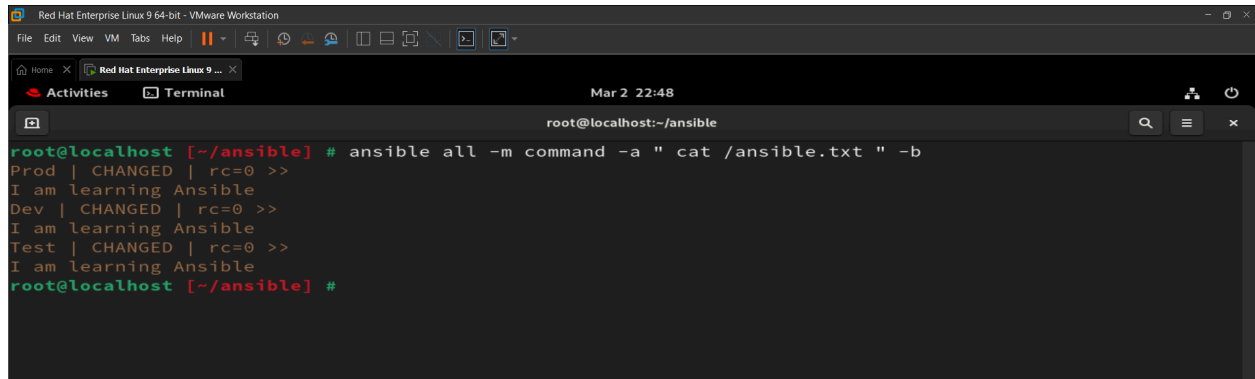


```

root@localhost [~/ansible] # ansible all -m lineinfile -a "path=/ansible.txt line='I am learning Ansible'
" -b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "backup": "",
  "changed": true,
  "msg": "line added"
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "backup": "",
  "changed": true,
  "msg": "line added"
}
Test | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "backup": "",
  "changed": true,
  "msg": "line added"
}
root@localhost [~/ansible] #

```

- Verify the line "I am learning Ansible" in the ansible.txt file.



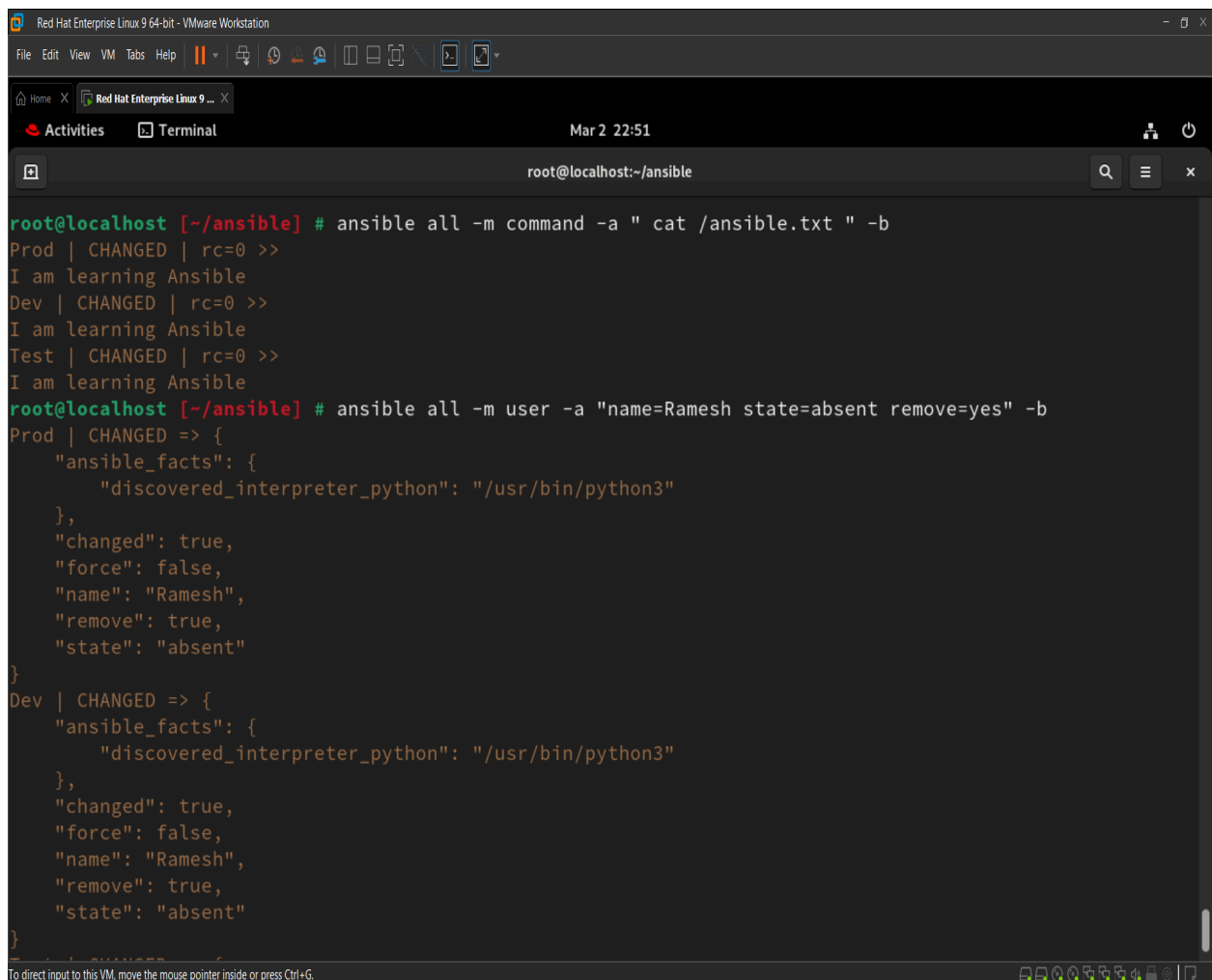
A terminal window titled "Red Hat Enterprise Linux 9 64-bit - VMware Workstation" showing the execution of an Ansible command. The prompt is "root@localhost:~/ansible". The command entered is "ansible all -m command -a 'cat /ansible.txt' -b". The output shows the command being executed on three hosts: Prod, Dev, and Test. Each host returns the content of the file /ansible.txt, which is "I am learning Ansible".

```
root@localhost [~/ansible] # ansible all -m command -a " cat /ansible.txt " -b
Prod | CHANGED | rc=0 >>
I am learning Ansible
Dev | CHANGED | rc=0 >>
I am learning Ansible
Test | CHANGED | rc=0 >>
I am learning Ansible
root@localhost [~/ansible] #
```

Q4. User Deletion and New User Creation:

- Delete the previously created users (Ramesh, Priya, Suresh).

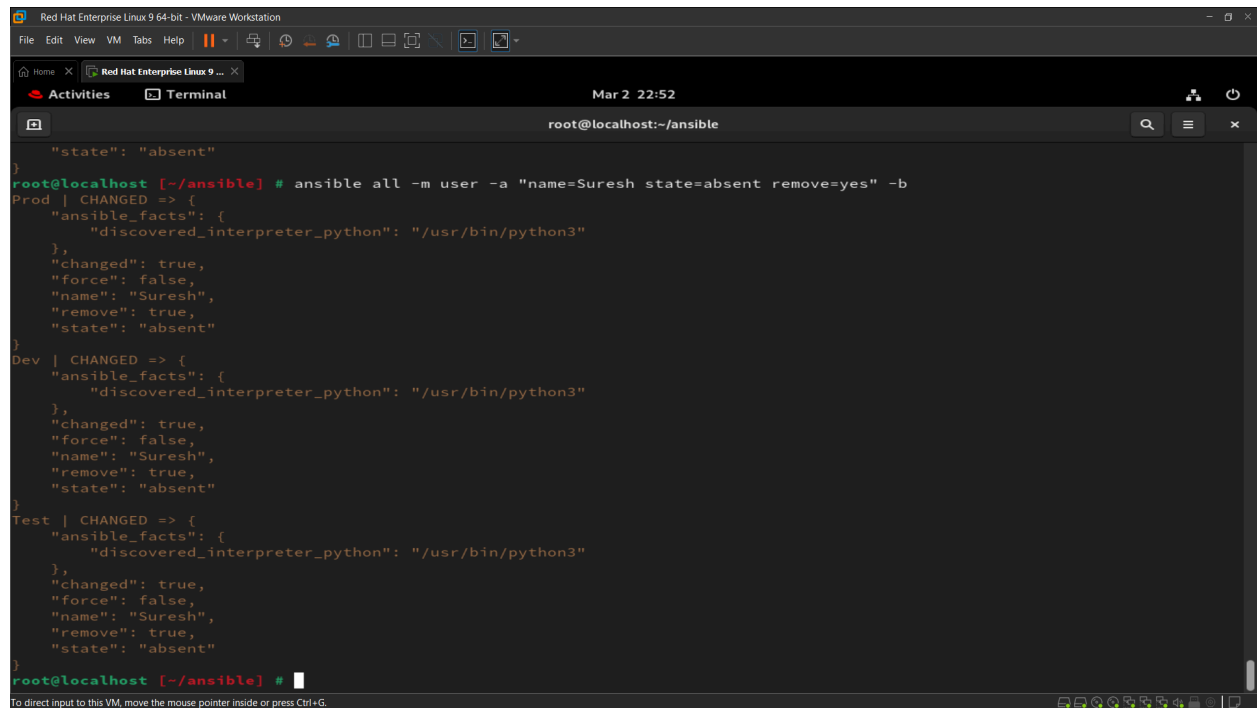
Ramesh :



A terminal window titled "Red Hat Enterprise Linux 9 64-bit - VMware Workstation" showing the execution of an Ansible command to delete the user Ramesh. The prompt is "root@localhost:~/ansible". The command entered is "ansible all -m user -a 'name=Ramesh state=absent remove=yes' -b". The output shows the command being executed on three hosts: Prod, Dev, and Test. Each host returns a JSON object indicating the user was successfully removed.

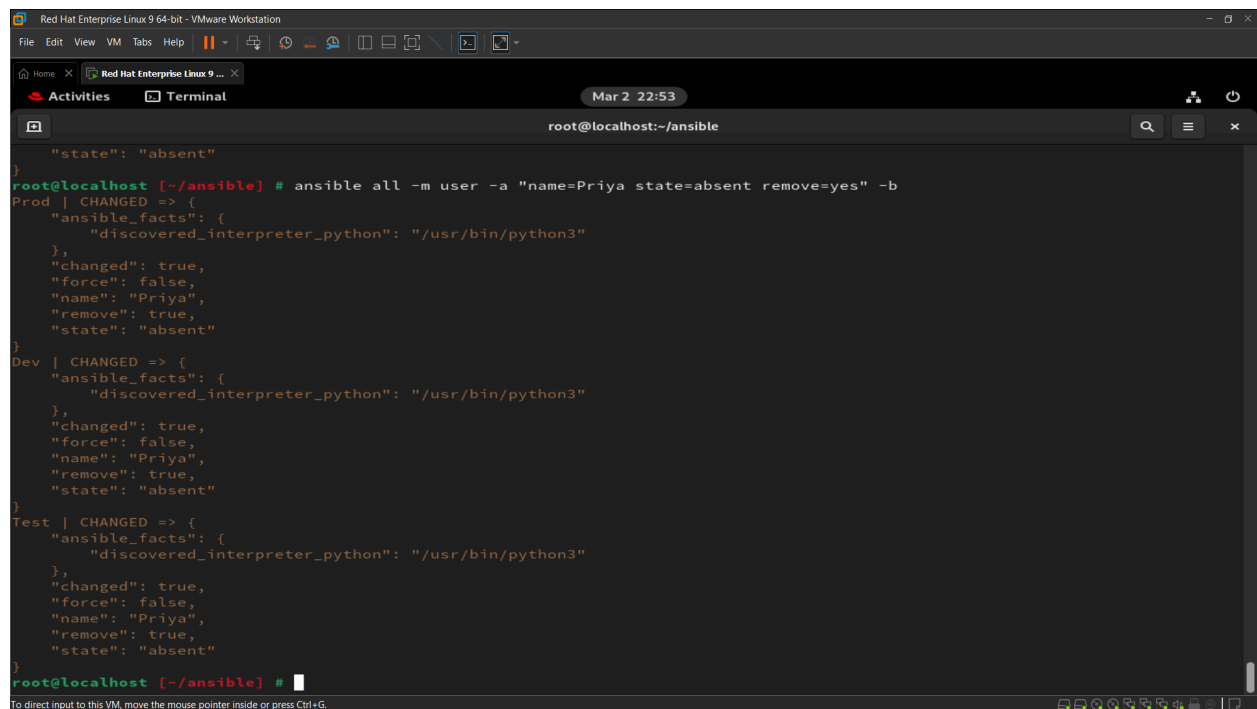
```
root@localhost [~/ansible] # ansible all -m user -a "name=Ramesh state=absent remove=yes" -b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "force": false,
  "name": "Ramesh",
  "remove": true,
  "state": "absent"
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "force": false,
  "name": "Ramesh",
  "remove": true,
  "state": "absent"
}
Test | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "force": false,
  "name": "Ramesh",
  "remove": true,
  "state": "absent"
}
```

Suresh :



```
root@localhost:~/ansible
}
"state": "absent"
root@localhost [~/ansible] # ansible all -m user -a "name=Suresh state=absent remove=yes" -b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "force": false,
  "name": "Suresh",
  "remove": true,
  "state": "absent"
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "force": false,
  "name": "Suresh",
  "remove": true,
  "state": "absent"
}
Test | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "force": false,
  "name": "Suresh",
  "remove": true,
  "state": "absent"
}
root@localhost [~/ansible] #
```

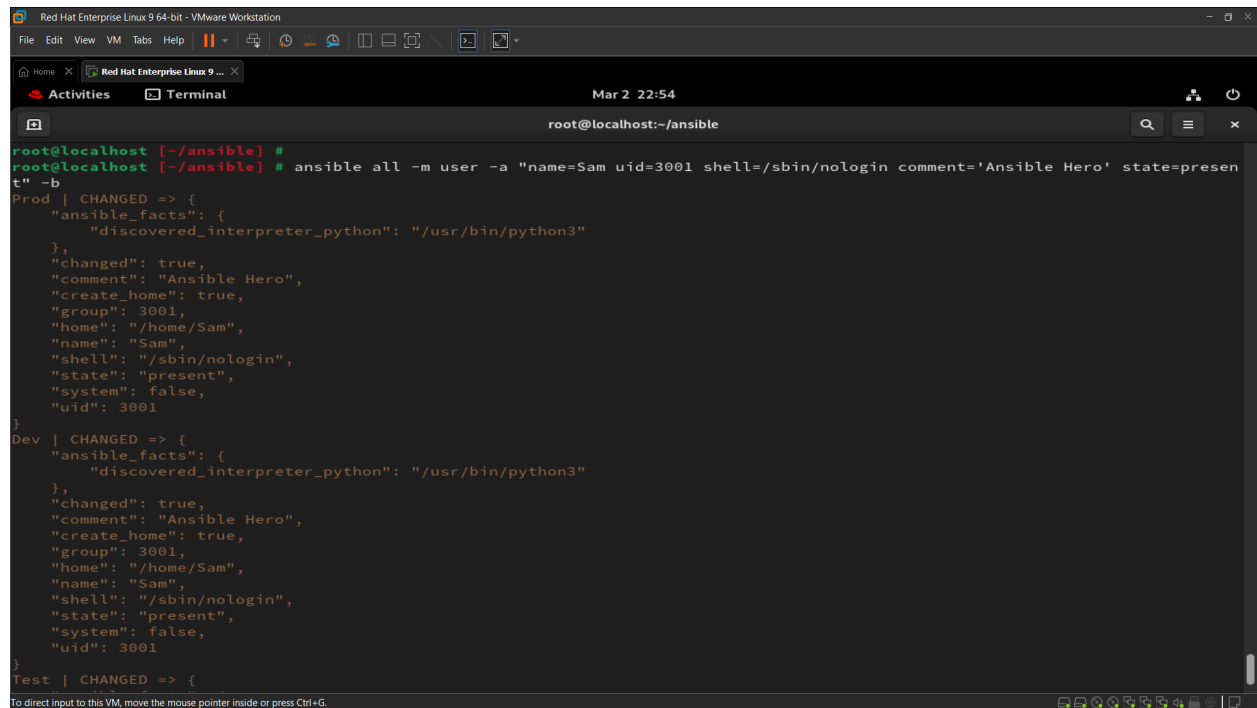
Priya:



```
root@localhost:~/ansible
}
"state": "absent"
root@localhost [~/ansible] # ansible all -m user -a "name=Priya state=absent remove=yes" -b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "force": false,
  "name": "Priya",
  "remove": true,
  "state": "absent"
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "force": false,
  "name": "Priya",
  "remove": true,
  "state": "absent"
}
Test | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "force": false,
  "name": "Priya",
  "remove": true,
  "state": "absent"
}
root@localhost [~/ansible] #
```

- Create new users:
- Add the comment "Ansible Hero" to all these new users.

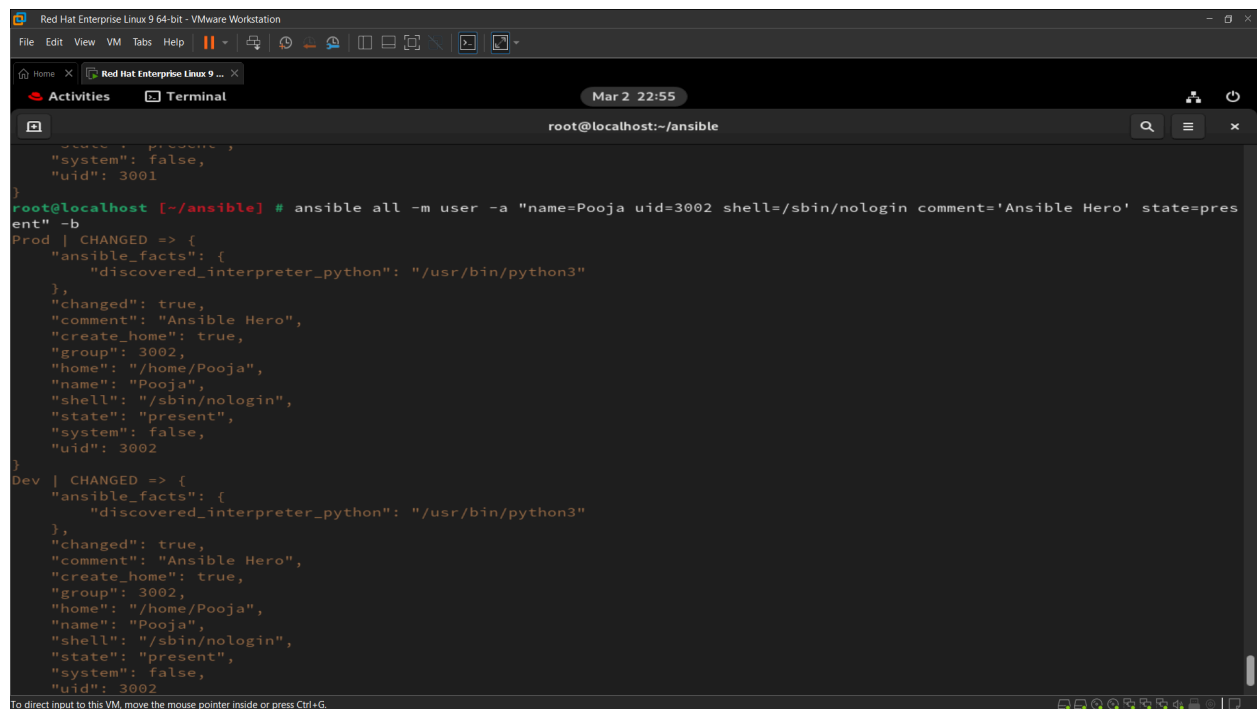
- Sam (UID: 3001, shell: /sbin/nologin)



```
Red Hat Enterprise Linux 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
Activities Terminal Mar 2 22:54
root@localhost:~/ansible

root@localhost [~/ansible] #
root@localhost [~/ansible] # ansible all -m user -a "name=Sam uid=3001 shell=/sbin/nologin comment='Ansible Hero' state=present" -b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "comment": "Ansible Hero",
  "create_home": true,
  "group": 3001,
  "home": "/home/Sam",
  "name": "Sam",
  "shell": "/sbin/nologin",
  "state": "present",
  "system": false,
  "uid": 3001
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "comment": "Ansible Hero",
  "create_home": true,
  "group": 3001,
  "home": "/home/Sam",
  "name": "Sam",
  "shell": "/sbin/nologin",
  "state": "present",
  "system": false,
  "uid": 3001
}
Test | CHANGED => {
```

- Pooja (UID: 3002, shell: /sbin/nologin)



```
Red Hat Enterprise Linux 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
Activities Terminal Mar 2 22:55
root@localhost:~/ansible

root@localhost [~/ansible] #
root@localhost [~/ansible] # ansible all -m user -a "name=Pooja uid=3002 shell=/sbin/nologin comment='Ansible Hero' state=present" -b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "comment": "Ansible Hero",
  "create_home": true,
  "group": 3002,
  "home": "/home/Pooja",
  "name": "Pooja",
  "shell": "/sbin/nologin",
  "state": "present",
  "system": false,
  "uid": 3002
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "comment": "Ansible Hero",
  "create_home": true,
  "group": 3002,
  "home": "/home/Pooja",
  "name": "Pooja",
  "shell": "/sbin/nologin",
  "state": "present",
  "system": false,
  "uid": 3002
}
Test | CHANGED => {
```

- Nike (UID: 3003)

```
Red Hat Enterprise Linux 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
Activities Terminal
root@localhost:~/ansible

root@localhost [~/ansible] # ansible all -m user -a "name=Nike uid=3003 shell=/sbin/nologin comment='Ansible Hero' state=present" -b
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "comment": "Ansible Hero",
  "create_home": true,
  "group": 3003,
  "home": "/home/Nike",
  "name": "Nike",
  "shell": "/sbin/nologin",
  "state": "present",
  "system": false,
  "uid": 3003
}
Dev | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "comment": "Ansible Hero",
  "create_home": true,
  "group": 3003,
  "home": "/home/Nike",
  "name": "Nike",
  "shell": "/sbin/nologin",
  "state": "present",
  "system": false,
  "uid": 3003
}
Test | CHANGED => {
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

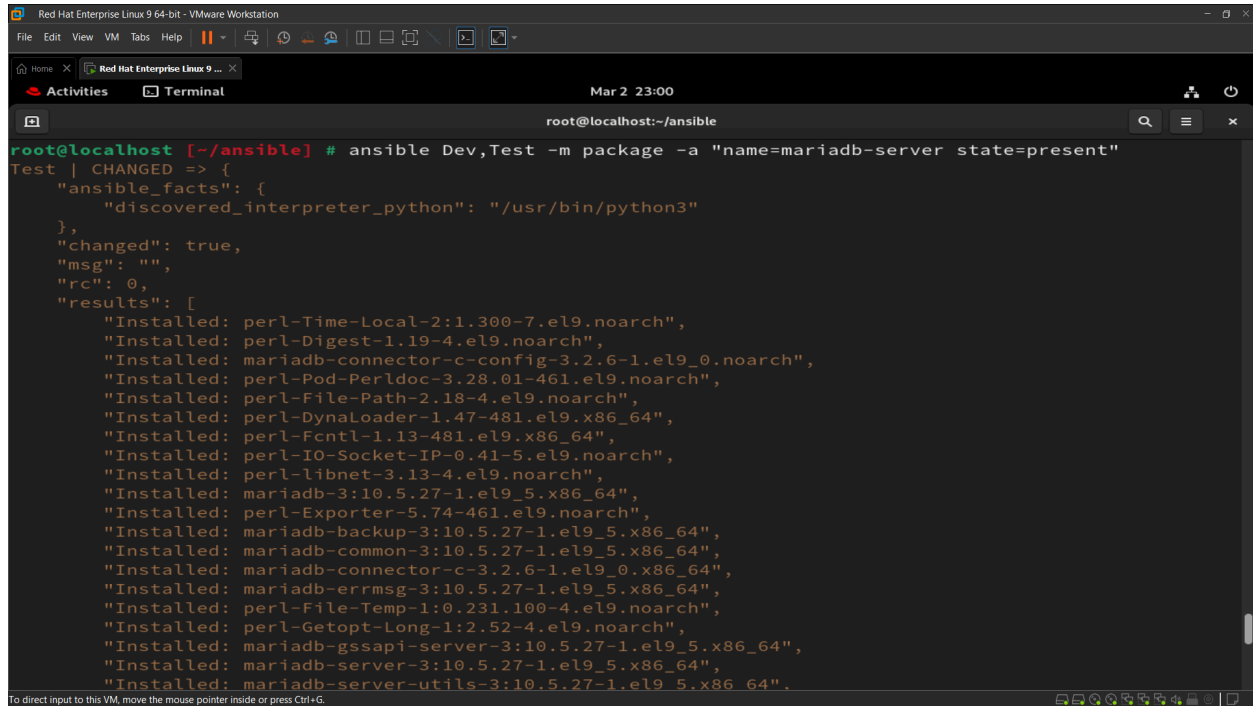
- Verify old user delete and new user create with uid,shell,comment

```
Red Hat Enterprise Linux 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
Activities Terminal
root@localhost:~/ansible

root@localhost [~/ansible] # ansible all -m command -a "id Sam "
Dev | CHANGED | rc=0 >>
uid=3001(Sam) gid=3001(Sam) groups=3001(Sam)
Prod | CHANGED | rc=0 >>
uid=3001(Sam) gid=3001(Sam) groups=3001(Sam)
Test | CHANGED | rc=0 >>
uid=3001(Sam) gid=3001(Sam) groups=3001(Sam)
root@localhost [~/ansible] # ansible all -m command -a "id Pooja "
Dev | CHANGED | rc=0 >>
uid=3002(Pooja) gid=3002(Pooja) groups=3002(Pooja)
Prod | CHANGED | rc=0 >>
uid=3002(Pooja) gid=3002(Pooja) groups=3002(Pooja)
Test | CHANGED | rc=0 >>
uid=3002(Pooja) gid=3002(Pooja) groups=3002(Pooja)
root@localhost [~/ansible] # ansible all -m command -a "id Nike "
Dev | CHANGED | rc=0 >>
uid=3003(Nike) gid=3003(Nike) groups=3003(Nike)
Prod | CHANGED | rc=0 >>
uid=3003(Nike) gid=3003(Nike) groups=3003(Nike)
Test | CHANGED | rc=0 >>
uid=3003(Nike) gid=3003(Nike) groups=3003(Nike)
root@localhost [~/ansible] # ansible all -m command -a "id Ramesh "
Dev | FAILED | rc=1 >>
id: 'Ramesh': no such usernon-zero return code
Prod | FAILED | rc=1 >>
id: 'Ramesh': no such usernon-zero return code
Test | FAILED | rc=1 >>
id: 'Ramesh': no such usernon-zero return code
root@localhost [~/ansible] # ansible all -m command -a "id Suresh "
Dev | FAILED | rc=1 >>
id: 'Suresh': no such usernon-zero return code
Prod | FAILED | rc=1 >>
id: 'Suresh': no such usernon-zero return code
Test | FAILED | rc=1 >>
id: 'Suresh': no such usernon-zero return code
root@localhost [~/ansible] # ansible all -m command -a "id Priya "
Dev | FAILED | rc=1 >>
id: 'Priya': no such usernon-zero return code
Prod | FAILED | rc=1 >>
id: 'Priya': no such usernon-zero return code
Test | FAILED | rc=1 >>
id: 'Priya': no such usernon-zero return code
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

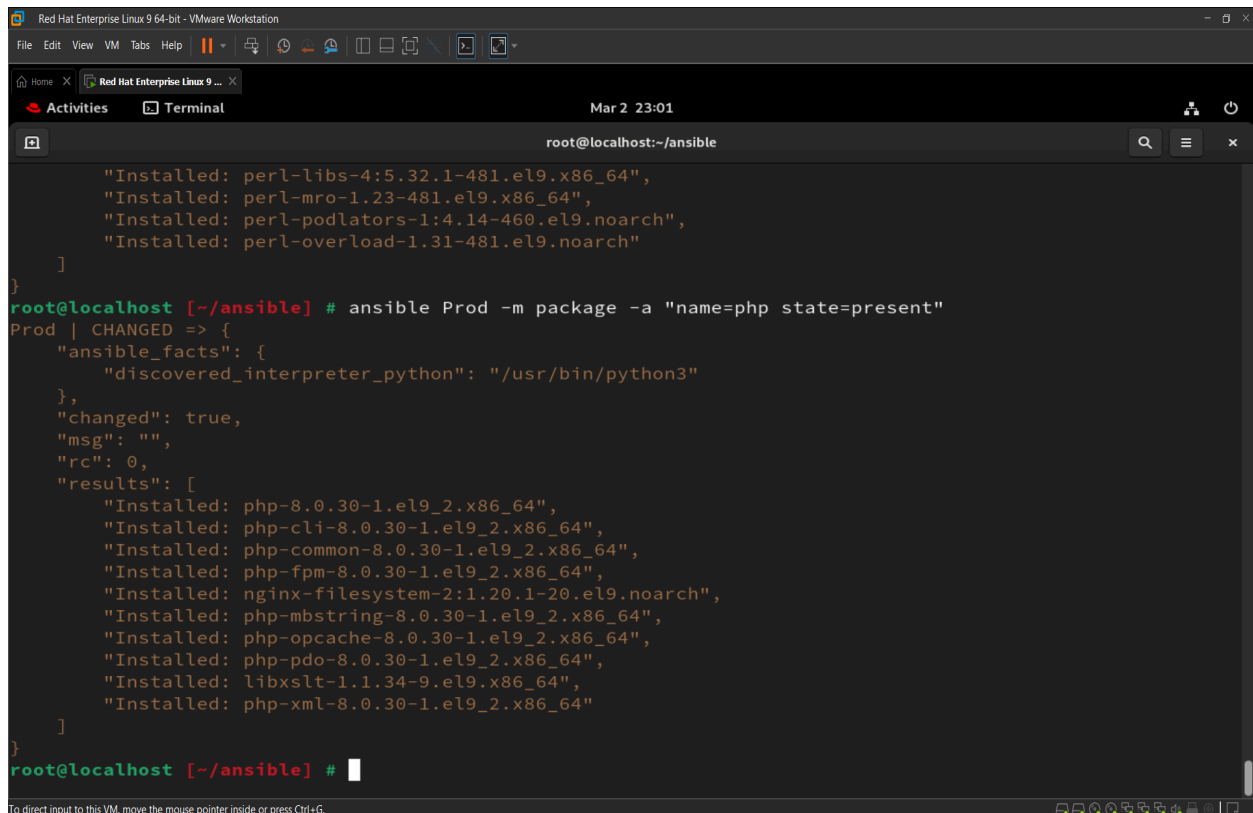
Q5. Software Installation:

- Install mariadb on the Dev and Test managed hosts.



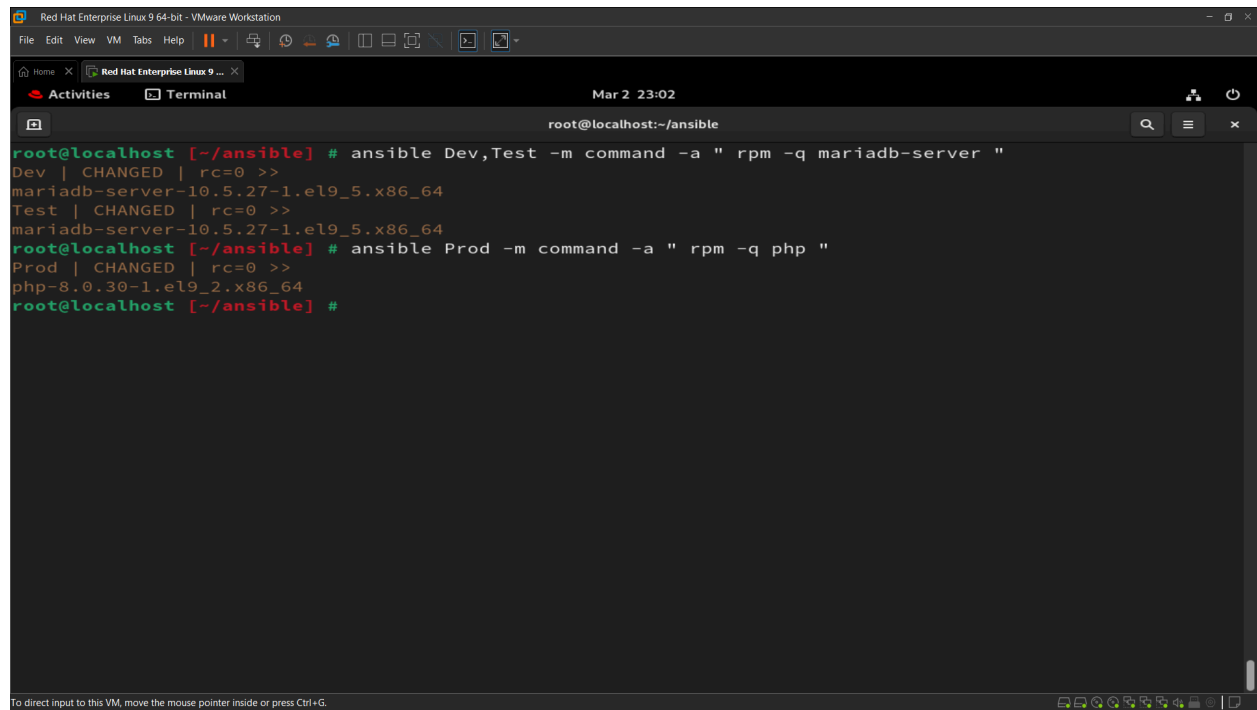
```
root@localhost:~/ansible
root@localhost [~/ansible] # ansible Dev,Test -m package -a "name=mariadb-server state=present"
Test | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "msg": "",
  "rc": 0,
  "results": [
    "Installed: perl-Time-Local-2:1.300-7.el9.noarch",
    "Installed: perl-Digest-1.19-4.el9.noarch",
    "Installed: mariadb-connector-c-config-3.2.6-1.el9_0.noarch",
    "Installed: perl-Pod-Perldoc-3.28.01-461.el9.noarch",
    "Installed: perl-File-Path-2.18-4.el9.noarch",
    "Installed: perl-DynaLoader-1.47-481.el9.x86_64",
    "Installed: perl-Fcntl-1.13-481.el9.x86_64",
    "Installed: perl-IO-Socket-IP-0.41-5.el9.noarch",
    "Installed: perl-libnet-3.13-4.el9.noarch",
    "Installed: mariadb-3:10.5.27-1.el9_5.x86_64",
    "Installed: perl-Exporter-5.74-461.el9.noarch",
    "Installed: mariadb-backup-3:10.5.27-1.el9_5.x86_64",
    "Installed: mariadb-common-3:10.5.27-1.el9_5.x86_64",
    "Installed: mariadb-connector-c-3.2.6-1.el9_0.x86_64",
    "Installed: mariadb-errmsg-3:10.5.27-1.el9_5.x86_64",
    "Installed: perl-File-Temp-1:0.231.100-4.el9.noarch",
    "Installed: perl-Getopt-Long-1:2.52-4.el9.noarch",
    "Installed: mariadb-gssapi-server-3:10.5.27-1.el9_5.x86_64",
    "Installed: mariadb-server-3:10.5.27-1.el9_5.x86_64",
    "Installed: mariadb-server-utils-3:10.5.27-1.el9_5.x86_64",
```

- Install php on the Prod managed host.



```
root@localhost:~/ansible
root@localhost [~/ansible] # ansible Prod -m package -a "name=php state=present"
Prod | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "msg": "",
  "rc": 0,
  "results": [
    "Installed: perl-libs-4:5.32.1-481.el9.x86_64",
    "Installed: perl-mro-1.23-481.el9.x86_64",
    "Installed: perl-podlators-1:4.14-460.el9.noarch",
    "Installed: perl-overload-1.31-481.el9.noarch"
  ]
}
root@localhost [~/ansible] #
```

- **Verify Packages.**



```
Red Hat Enterprise Linux 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
Activities Terminal Mar 2 23:02
root@localhost:~/ansible

root@localhost [~/ansible] # ansible Dev,Test -m command -a " rpm -q mariadb-server "
Dev | CHANGED | rc=0 >>
mariadb-server-10.5.27-1.el9_5.x86_64
Test | CHANGED | rc=0 >>
mariadb-server-10.5.27-1.el9_5.x86_64
root@localhost [~/ansible] # ansible Prod -m command -a " rpm -q php "
Prod | CHANGED | rc=0 >>
php-8.0.30-1.el9_2.x86_64
root@localhost [~/ansible] #
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.